

Cleavage sites, scores and mechanisms

Sid's disclosed synthetic long vaccine peptides | human study | frozen local predictions

Coverage: 40 source records, 39 unique SLP sequences. RNA-encoded sequences are outside this view. Gold marks the source-listed minimal epitope only where its recorded offset matches. That annotation is an mRNA minimal/candidate epitope and does not establish the experimentally used minimal target of every SLP.

Coordinates: bond b is between SLP residues b and b+1. An internal core cut lies strictly between the core boundaries. Each numeric cell is one model's native score, rounded to three decimals for display; CSV retains the original precision. NA means unassessed/unsupported, never zero. M/N means motif matched/not matched, never a numerical score.

Mechanism: ligand-trained models report processing proxies; they cannot name the enzyme responsible for a bond. 20S reports an in-vitro proteasome model. Named peptidases have their own assay/motif evidence. For injected SLPs, cytosolic scores require uptake/export, ER trimming requires ER access, and extracellular activity requires enzyme exposure. No uptake, successive digestion, kinetics, MHC protection, or overall survival is predicted.

Pepsickle preference: organism context = human; Pepsickle human is preferred. All-mammal is preferred for nonhuman, mixed or unknown context. Both scores remain separate; this is a user-selected population policy, not evidence that human-only is more accurate. Upstream calls human-only experimental and notes its smaller training set. All-mammal training does not validate arbitrary non-mammalian organisms.

NetChop performance: DTU reports Cterm 3.0 performs best for CTL epitope boundaries. It is trained on MHC-I ligand C-termini; 20S uses in-vitro degradation data. Cterm is useful for ligand boundaries, while 20S has the more direct purified-proteasome interpretation. Neither score is a clinical cleavage probability. The 0.5 emphasis applies only to each 0-1 processing row; it is not a cross-model calibration.

Provenance: source checksums verified; inference not repeated; no CleaveNet results included. Full model identities, assay scopes and primary references are in mechanism_catalog.csv; exact states and scores are in cleavage_sites.csv. The three exact-sequence reference models (THOP1, NLN, IRAP) were excluded in the source run and are not extrapolated here.

Processing models: distinct endpoints

The same bond can score differently because training endpoints differ; no ensemble score is computed.

Model	Biological route / enzyme attribution	Training endpoint and applicability
Pepsickle human (preferred)	MHC-I processing proxy; proteasome-associated; type agnostic	Human epitope-trained neural ensemble; no constitutive/immunoproteasome assignment.
Pepsickle all-mammal	MHC-I processing proxy; proteasome-associated; type agnostic	Mammalian epitope-trained neural ensemble; no constitutive/immunoproteasome assignment.
NetChop Cterm 3.0	MHC-I processing proxy; no individual enzyme assigned	Trained on MHC-I ligand C-termini; processing/selection signal, not a purified-enzyme assay.
NetChop 20S 3.0	Proteasome digestion; 20S proteasome; no catalytic subunit assigned	Trained on in-vitro proteasome degradation experiments.
NetCleave-I HLA	MHC-I processing proxy; no individual enzyme assigned	MHC-I ligand-derived C-terminal processing; 8 upstream plus 3 downstream residues in this run.
NetCleave-II HLA	MHC-II processing proxy; no named cathepsin assigned	MHC-II ligand-derived C-terminal processing; 13 upstream plus 3 downstream residues in this run.

DPP4 qPISA: only the exposed N-terminal dipeptide bond (b=2) of the intact SLP. Native log2 substrate depletion, not 0-1 cleavage probability or serum half-life. Missing coefficients cause abstention. **ERAMER:** ERAP1's initial N-terminal trimming step (b=1), only for 9-16-residue inputs; native PWM specificity.

Key primary sources:

NetChop: <https://services.healthtech.dtu.dk/services/NetChop-3.1/> ; DOI 10.1007/s00251-005-0781-7

Pepsickle: DOI 10.1093/bioinformatics/btab628 ; <https://github.com/pdngx/pepsickle>

NetCleave: DOI 10.1038/s41598-021-92632-y

DPP4: DOI 10.1038/s44320-024-00071-4 ; ERAMER: PMID 38925438

Named peptidases: topology matters

All assessments use the intact SLP with assumed free termini; repeated trimming and newly exposed fragments are not simulated.

Model / enzyme	Topology	Annotated locations	Evidence
dpp4-qpisa / DPP4	n terminal; removes 2	extracellular;plasma;serum	log2 fold change (buffer control / DPP4-treated)
app1-xp / XPNPEP1	n terminal; removes 1	cytosol	required motif
tpp2-tripeptidyl / TPP2	n terminal; removes 3	cytosol	permissive motif
npepps-n-terminal / NPEPPS	n terminal; removes 1	cytosol	permissive motif
ace-dipeptidyl / ACE	c terminal; removes 2	plasma;serum;extracellular	required motif
mme-hydrophobic / MME	internal	extracellular	preferred motif
cpb2-basic / CPB2	c terminal; removes 1	plasma;serum;extracellular	required motif
cpn-basic / CPN1	c terminal; removes 1	plasma;serum;extracellular	required motif
app2-xp / XPNPEP2	n terminal; removes 1	plasma;extracellular	required motif
fap-dipeptidyl / FAP	n terminal; removes 2	extracellular;plasma	required motif
fap-endo-gp / FAP	internal	extracellular;plasma	required motif
enpep-acidic / ENPEP	n terminal; removes 1	extracellular	preferred motif
anpep-ala / ANPEP	n terminal; removes 1	extracellular	permissive motif
dpp8-xp-xa / DPP8	n terminal; removes 2	cytosol	required motif
dpp9-xp-xa / DPP9	n terminal; removes 2	cytosol	required motif
prep-pro / PREP	internal	cytosol;extracellular;serum	required motif
erap2-basic / ERAP2	n terminal; removes 1	er	preferred motif
eramer-step / ERAP1	n terminal; removes 1	er	native ERAMER specificity

Required = clears this rule's recognition gate; preferred = incomplete preference; permissive = broad, weak flag. None establishes a cleavage rate. PREP: PMID 22750443; MME: PMID 6349683; FAP: PMID 16480718. CPB2 explicitly assumes activated enzyme in the frozen analysis. Every rule's complete primary references and limitations are in the catalog.

ADGRF5 p.Asn1279Ser

JLF V2 | 22 aa | bonds 1-21 | segment 1/1

VQEALLSKFSLSRWSSQHSKST

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
	V Q	Q E	E A	A L	L L	L S	S K	K F	F S	S L	L S	S R	R W	W S	S S	S Q	Q H	H S	S K	K S	S T
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.225	0.013	0.009	0.007	0.013	0.022	0.016	0.661	0.902	0.148	0.827	0.127	0.428	0.732	0.111	0.085	0.146	0.186	0.173	0.894	0.231
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.001	0.007	0.007	0.004	0.012	0.024	0.027	0.672	0.830	0.133	0.782	0.130	0.356	0.757	0.072	0.066	0.148	0.184	0.249	0.841	0.357
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.698	0.060	0.037	0.733	0.619	0.499	0.033	0.796	0.856	0.023	0.929	0.195	0.962	0.912	0.029	0.077	0.068	0.831	0.214	0.976	0.240
NetChop 20S 3.0 In-vitro proteasome digestion model	0.213	0.518	0.953	0.477	0.126	0.181	0.730	0.174	0.853	0.189	0.583	0.687	0.812	0.508	0.227	0.868	0.174	0.165	0.061	0.273	0.117
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.200	0.755	0.013	0.847	0.037	0.153	0.822	0.012	0.246	0.149	0.545	0.031	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.393	0.296	0.394	0.261	0.358	0.381	0.238	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	M	M	N	N	M	N	M	N	N	N	N	N	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa Q2 E3 : 0.019 log2 depletion	FAP [N2] / fap-dipeptidyl Q2 E3 : N
XPNPEP1 [N1] / app1-xp V1 Q2 : N	ENPEP [N1] / enpep-acidic V1 Q2 : N
TPP2 [N3] / tpp2-tripeptidyl E3 A4 : M	ANPEP [N1] / anpep-ala V1 Q2 : N
NPEPPS [N1] / npepps-n-terminal V1 Q2 : M	DPP8 [N2] / dpp8-xp-xa Q2 E3 : N
ACE [C2] / ace-dipeptidyl K20 S21 : M	DPP9 [N2] / dpp9-xp-xa Q2 E3 : N
CPB2 (active assumption) [C1] / cpb2-basic S21 T22 : N	ERAP2 [N1] / erap2-basic V1 Q2 : N
CPN1 [C1] / cpn-basic S21 T22 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp V1 Q2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

BMP1 p.Ser366Thr

JLF V3 | 23 aa | bonds 1-22 | segment 1/1

RISVTPGEKIILNFTTLDLYRSR

Source minimal epitope: KIILNFTTL (SLP 9-17)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
	R I	I S	S V	V T	T P	P G	G E	E K	K I	I I	I L	L N	N F	F T	T T	T L	L D	D L	L Y	Y R	R S	S R
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.317	0.005	0.009	0.006	0.020	0.002	0.004	0.158	0.549	0.456	0.429	0.814	0.228	0.825	0.123	0.131	0.659	0.192	0.704	0.802	0.692	0.451
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.005	0.002	0.005	0.005	0.007	0.006	0.027	0.059	0.573	0.452	0.364	0.734	0.155	0.572	0.157	0.097	0.714	0.131	0.668	0.843	0.639	0.505
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.167	0.763	0.024	0.424	0.048	0.051	0.025	0.031	0.287	0.142	0.403	0.891	0.234	0.361	0.043	0.030	0.970	0.259	0.976	0.812	0.570	0.239
NetChop 20S 3.0 In-vitro proteasome digestion model	0.274	0.525	0.238	0.812	0.291	0.056	0.027	0.713	0.271	0.420	0.835	0.963	0.384	0.988	0.295	0.553	0.948	0.872	0.730	0.721	0.833	0.503
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.182	0.369	0.538	0.079	0.758	0.047	0.891	0.069	0.038	0.941	0.026	0.218	0.625	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.280	0.518	0.231	0.230	0.395	0.263	0.537	0.298	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	M	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	M	N	N	N	N	N	N	N	M	M	M	N	M	N	N	M	N	M	M	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa I2 S3 : -0.005 log2 depletion	FAP [N2] / fap-dipeptidyl I2 S3 : N
XPNPEP1 [N1] / app1-xp R1 I2 : N	ENPEP [N1] / enpep-acidic R1 I2 : N
TPP2 [N3] / tpp2-tripeptidyl S3 V4 : M	ANPEP [N1] / anpep-ala R1 I2 : N
NPEPPS [N1] / npepps-n-terminal R1 I2 : M	DPP8 [N2] / dpp8-xp-xa I2 S3 : N
ACE [C2] / ace-dipeptidyl R21 S22 : M	DPP9 [N2] / dpp9-xp-xa I2 S3 : N
CPB2 (active assumption) [C1] / cpb2-basic S22 R23 : M	ERAP2 [N1] / erap2-basic R1 I2 : M
CPN1 [C1] / cpn-basic S22 R23 : M	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp R1 I2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

BTD p.Phe445Val

JLF V3;JLF V2;JLF V1 | 23 aa | bonds 1-22 | segment 1/1

RPTLSKELYALGVVDGLHTVHGT

Source minimal epitope: GVVDGLHTV (SLP 12-20)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
	R P	P T	T L	L S	S K	K E	E L	L Y	Y A	A L	L G	G V	V V	V D	D G	G L	L H	H T	T V	V H	H G	G T
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.072	0.003	0.011	0.013	0.008	0.035	0.037	0.722	0.939	0.343	0.778	0.122	0.277	0.255	0.096	0.157	0.597	0.172	0.281	0.929	0.801	0.468
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.003	0.002	0.005	0.009	0.017	0.022	0.061	0.583	0.897	0.410	0.743	0.108	0.307	0.248	0.087	0.097	0.525	0.180	0.243	0.918	0.703	0.420
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.150	0.075	0.029	0.900	0.039	0.916	0.030	0.940	0.957	0.076	0.923	0.050	0.168	0.323	0.079	0.059	0.925	0.064	0.040	0.977	0.784	0.220
NetChop 20S 3.0 In-vitro proteasome digestion model	0.593	0.112	0.027	0.426	0.529	0.716	0.358	0.937	0.895	0.703	0.729	0.079	0.780	0.522	0.909	0.442	0.734	0.176	0.634	0.708	0.343	0.452
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.218	0.706	0.056	0.780	0.028	0.687	0.333	0.270	0.049	0.876	0.145	0.039	0.921	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.184	0.258	0.407	0.411	0.260	0.333	0.316	0.245	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	M	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	M	N	N	N	M	M	N	M	N	N	N	N	N	M	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa P2 T3 : 3.148 log2 depletion	FAP [N2] / fap-dipeptidyl P2 T3 : M
XPNPEP1 [N1] / app1-xp R1 P2 : M	ENPEP [N1] / enpep-acidic R1 P2 : N
TPP2 [N3] / tpp2-tripeptidyl T3 L4 : M	ANPEP [N1] / anpep-ala R1 P2 : N
NPEPPS [N1] / npepps-n-terminal R1 P2 : N	DPP8 [N2] / dpp8-xp-xa P2 T3 : M
ACE [C2] / ace-dipeptidyl H21 G22 : M	DPP9 [N2] / dpp9-xp-xa P2 T3 : M
CPB2 (active assumption) [C1] / cpb2-basic G22 T23 : N	ERAP2 [N1] / erap2-basic R1 P2 : M
CPN1 [C1] / cpn-basic G22 T23 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp R1 P2 : M	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

CD109 p.Arg1310Met

JLF V3 | 23 aa | bonds 1-22 | segment 1/1

SFSGPGMSGMALMEVNLLSGKKK

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
	S F	F S	S G	G P	P G	G M	M S	S G	G M	M A	A L	L M	M E	E V	V N	N L	L L	L S	S G	G K	K K	K K
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.017	0.032	0.006	0.006	0.004	0.004	0.093	0.057	0.146	0.790	0.139	0.609	0.329	0.070	0.593	0.094	0.502	0.687	0.144	0.309	0.849	0.912
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.009	0.011	0.008	0.009	0.003	0.013	0.119	0.047	0.203	0.689	0.171	0.619	0.293	0.085	0.439	0.101	0.488	0.632	0.161	0.366	0.794	0.906
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.048	0.837	0.024	0.049	0.026	0.041	0.294	0.031	0.081	0.252	0.083	0.712	0.576	0.024	0.499	0.024	0.630	0.848	0.220	0.184	0.736	0.933
NetChop 20S 3.0 In-vitro proteasome digestion model	0.086	0.765	0.066	0.500	0.040	0.027	0.836	0.146	0.727	0.926	0.273	0.914	0.681	0.953	0.935	0.284	0.692	0.697	0.037	0.051	0.034	0.066
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.036	0.132	0.802	0.072	0.750	0.313	0.034	0.923	0.035	0.258	0.679	0.026	0.095	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.169	0.327	0.195	0.301	0.320	0.283	0.242	0.307	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	M	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	M	N	N	N	N	N	N	N	N	N	M	N	N	N	N	M	M	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	M	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa F2 S3 : 0.016 log2 depletion	FAP [N2] / fap-dipeptidyl F2 S3 : N
XPNPEP1 [N1] / app1-xp S1 F2 : N	ENPEP [N1] / enpep-acidic S1 F2 : N
TPP2 [N3] / tpp2-tripeptidyl S3 G4 : M	ANPEP [N1] / anpep-ala S1 F2 : N
NPEPPS [N1] / npepps-n-terminal S1 F2 : M	DPP8 [N2] / dpp8-xp-xa F2 S3 : N
ACE [C2] / ace-dipeptidyl K21 K22 : M	DPP9 [N2] / dpp9-xp-xa F2 S3 : N
CPB2 (active assumption) [C1] / cpb2-basic K22 K23 : M	ERAP2 [N1] / erap2-basic S1 F2 : N
CPN1 [C1] / cpn-basic K22 K23 : M	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp S1 F2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

CDC40 p.Trp475Ser

JLF V3 | 25 aa | bonds 1-24 | segment 1/1

HSMPAVTLSPNGKSLACQSM DNQKK

Source minimal epitope: TLSPNGKSL (SLP 7-15)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
	H S	S M	M P	P A	A V	V T	T L	L S	S P	P N	N G	G K	K S	S L	L A	A C	C Q	Q S	S M	M D	D N	N Q	Q K	K K
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.019	0.007	0.016	0.003	0.008	0.018	0.032	0.863	0.096	0.138	0.232	0.098	0.367	0.108	0.911	0.590	0.798	0.287	0.173	0.539	0.271	0.115	0.239	0.785
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.017	0.001	0.008	0.004	0.012	0.009	0.037	0.748	0.090	0.154	0.229	0.118	0.456	0.092	0.905	0.527	0.747	0.362	0.168	0.474	0.170	0.155	0.295	0.765
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.061	0.035	0.799	0.068	0.330	0.591	0.030	0.966	0.030	0.418	0.065	0.034	0.908	0.126	0.973	0.833	0.178	0.029	0.030	0.770	0.032	0.141	0.034	0.322
NetChop 20S 3.0 In-vitro proteasome digestion model	0.366	0.100	0.591	0.046	0.083	0.678	0.114	0.901	0.482	0.140	0.026	0.132	0.242	0.051	0.946	0.225	0.418	0.739	0.682	0.658	0.400	0.038	0.194	0.109
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.960	0.020	0.582	0.403	0.070	0.893	0.018	0.888	0.098	0.189	0.300	0.066	0.796	0.055	0.179	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.358	0.359	0.358	0.512	0.055	0.343	0.331	0.239	0.354	0.417	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	M	N	N	N	N	N	M	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	N	N	N	M	N	N	N	N	N	N	M	N	N	N	N	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa S2 M3 : 1.703 log2 depletion	FAP [N2] / fap-dipeptidyl S2 M3 : N
XPNPEP1 [N1] / app1-xp H1 S2 : N	ENPEP [N1] / enpep-acidic H1 S2 : N
TPP2 [N3] / tpp2-tripeptidyl M3 P4 : N	ANPEP [N1] / anpep-ala H1 S2 : N
NPEPPS [N1] / npepps-n-terminal H1 S2 : M	DPP8 [N2] / dpp8-xp-xa S2 M3 : N
ACE [C2] / ace-dipeptidyl Q23 K24 : M	DPP9 [N2] / dpp9-xp-xa S2 M3 : N
CPB2 (active assumption) [C1] / cpb2-basic K24 K25 : M	ERAP2 [N1] / erap2-basic H1 S2 : N
CPN1 [C1] / cpn-basic K24 K25 : M	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp H1 S2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

CENPL p.Pro26His

JLF V3 | 23 aa | bonds 1-22 | segment 1/1

SSRPEDYFIGATHLQKRLESVRK

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
	S S	S R	R P	P E	E D	D Y	Y F	F I	I G	G A	A T	T H	H L	L Q	Q K	K R	R L	L E	E S	S V	V R	R K
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.010	0.012	0.015	0.002	0.004	0.013	0.085	0.222	0.272	0.070	0.668	0.089	0.341	0.825	0.400	0.903	0.583	0.454	0.150	0.199	0.922	0.791
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.042	0.003	0.008	0.003	0.018	0.018	0.070	0.141	0.316	0.089	0.553	0.069	0.327	0.828	0.263	0.867	0.555	0.448	0.208	0.214	0.849	0.747
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.065	0.027	0.395	0.037	0.023	0.024	0.595	0.129	0.539	0.200	0.867	0.134	0.430	0.977	0.055	0.834	0.180	0.829	0.025	0.058	0.893	0.405
NetChop 20S 3.0 In-vitro proteasome digestion model	0.169	0.246	0.444	0.118	0.071	0.922	0.767	0.450	0.173	0.791	0.981	0.109	0.282	0.247	0.717	0.504	0.506	0.726	0.788	0.202	0.194	0.595
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.344	0.282	0.037	0.762	0.117	0.066	0.796	0.034	0.552	0.168	0.541	0.049	0.029	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.494	0.267	0.360	0.275	0.456	0.205	0.363	0.300	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	M	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	N	N	M	M	M	N	N	N	N	M	N	N	N	M	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa no assessed bond : NA	FAP [N2] / fap-dipeptidyl S2 R3 : N
XPNPEP1 [N1] / app1-xp S1 S2 : N	ENPEP [N1] / enpep-acidic S1 S2 : N
TPP2 [N3] / tpp2-tripeptidyl R3 P4 : N	ANPEP [N1] / anpep-ala S1 S2 : N
NPEPPS [N1] / npepps-n-terminal S1 S2 : M	DPP8 [N2] / dpp8-xp-xa S2 R3 : N
ACE [C2] / ace-dipeptidyl V21 R22 : M	DPP9 [N2] / dpp9-xp-xa S2 R3 : N
CPB2 (active assumption) [C1] / cpb2-basic R22 K23 : M	ERAP2 [N1] / erap2-basic S1 S2 : N
CPN1 [C1] / cpn-basic R22 K23 : M	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp S1 S2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

CUL9 p.Asp403Tyr

JLF V2 | 23 aa | bonds 1-22 | segment 1/1

RMLDYYEEISAGDEGEFRQSKKK

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
	R M	M L	L D	D Y	Y Y	Y E	E E	E I	I S	S A	A G	G D	D E	E G	G E	E F	F R	R Q	Q S	S K	K K	K K
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.469	0.010	0.006	0.013	0.009	0.018	0.012	0.220	0.954	0.102	0.442	0.041	0.076	0.070	0.104	0.143	0.621	0.468	0.201	0.439	0.878	0.904
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.022	0.001	0.002	0.030	0.008	0.018	0.043	0.150	0.907	0.117	0.456	0.052	0.129	0.062	0.048	0.161	0.460	0.410	0.179	0.345	0.881	0.909
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.455	0.440	0.661	0.029	0.803	0.363	0.027	0.104	0.958	0.064	0.050	0.360	0.137	0.023	0.032	0.024	0.776	0.139	0.306	0.111	0.889	0.861
NetChop 20S 3.0 In-vitro proteasome digestion model	0.593	0.817	0.958	0.938	0.786	0.781	0.697	0.275	0.631	0.199	0.436	0.060	0.647	0.063	0.509	0.101	0.489	0.389	0.072	0.083	0.044	0.081
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.083	0.904	0.030	0.782	0.076	0.339	0.307	0.239	0.083	0.684	0.276	0.029	0.167	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.134	0.364	0.339	0.175	0.449	0.322	0.307	0.375	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	M	N	M	M	N	N	M	N	N	N	N	N	N	N	M	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa no assessed bond : NA	FAP [N2] / fap-dipeptidyl M2 L3 : N
XPNPEP1 [N1] / app1-xp R1 M2 : N	ENPEP [N1] / enpep-acidic R1 M2 : N
TPP2 [N3] / tpp2-tripeptidyl L3 D4 : M	ANPEP [N1] / anpep-ala R1 M2 : N
NPEPPS [N1] / npepps-n-terminal R1 M2 : M	DPP8 [N2] / dpp8-xp-xa M2 L3 : N
ACE [C2] / ace-dipeptidyl K21 K22 : M	DPP9 [N2] / dpp9-xp-xa M2 L3 : N
CPB2 (active assumption) [C1] / cpb2-basic K22 K23 : M	ERAP2 [N1] / erap2-basic R1 M2 : M
CPN1 [C1] / cpn-basic K22 K23 : M	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp R1 M2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

DYNC1H1 p.Val314Ile

JLF V3;JLF V2 | 24 aa | bonds 1-23 | segment 1/1

GKRFHATISFDTDTGLKQALETKK

Source minimal epitope: KRFHATISF (SLP 2-10)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
	G K	K R	R F	F H	H A	A T	T I	I S	S F	F D	D T	T D	D T	T G	G L	L K	K Q	Q A	A L	L E	E T	T K	K K
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.007	0.045	0.011	0.015	0.006	0.045	0.028	0.288	0.105	0.922	0.130	0.061	0.072	0.031	0.077	0.654	0.543	0.232	0.360	0.728	0.210	0.534	0.955
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.029	0.006	0.008	0.006	0.007	0.016	0.049	0.280	0.095	0.911	0.113	0.049	0.097	0.047	0.083	0.507	0.515	0.276	0.342	0.700	0.238	0.390	0.918
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.044	0.318	0.382	0.533	0.032	0.484	0.035	0.962	0.030	0.975	0.026	0.085	0.031	0.061	0.043	0.970	0.833	0.178	0.557	0.919	0.033	0.177	0.961
NetChop 20S 3.0 In-vitro proteasome digestion model	0.152	0.499	0.579	0.854	0.909	0.857	0.102	0.846	0.298	0.932	0.876	0.163	0.949	0.047	0.053	0.485	0.084	0.863	0.599	0.496	0.058	0.083	0.180
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.904	0.021	0.936	0.042	0.245	0.412	0.385	0.046	0.555	0.249	0.084	0.153	0.833	0.029	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.421	0.296	0.411	0.193	0.406	0.357	0.372	0.293	0.340	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	M	N	N	N	M	N	M	N	N	N	N	N	M	N	N	N	M	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa | no assessed bond : NA

XPNPEP1 [N1] / app1-xp | G1|K2 : N

TPP2 [N3] / tpp2-tripeptidyl | R3|F4 : M

NPEPPS [N1] / npepps-n-terminal | G1|K2 : N

ACE [C2] / ace-dipeptidyl | T22|K23 : M

CPB2 (active assumption) [C1] / cpb2-basic | K23|K24 : M

CPN1 [C1] / cpn-basic | K23|K24 : M

XPNPEP2 [N1] / app2-xp | G1|K2 : N

FAP [N2] / fap-dipeptidyl | K2|R3 : N

ENPEP [N1] / enpep-acidic | G1|K2 : N

ANPEP [N1] / anpep-ala | G1|K2 : N

DPP8 [N2] / dpp8-xp-xa | K2|R3 : N

DPP9 [N2] / dpp9-xp-xa | K2|R3 : N

ERAP2 [N1] / erap2-basic | G1|K2 : N

ERAP1 [N1] / eramer-step | no assessed bond : NA

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

DYNC1H1 p.Val314Ile

CeGaT | 18 aa | bonds 1-17 | segment 1/1

KHGKRFHATISFDTDTGL

Source minimal epitope: KRFHATISF (SLP 4-12)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
	K H	H G	G K	K R	R F	F H	H A	A T	T I	I S	S F	F D	D T	T D	D T	T G	G L
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.433	0.007	0.006	0.013	0.005	0.006	0.011	0.108	0.116	0.445	0.173	0.934	0.229	0.107	0.197	0.145	0.246
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.004	0.004	0.003	0.011	0.008	0.006	0.058	0.084	0.118	0.515	0.114	0.911	0.166	0.113	0.228	0.193	0.338
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.694	0.091	0.051	0.400	0.567	0.339	0.031	0.445	0.044	0.969	0.035	0.975	0.031	0.129	0.070	0.256	0.227
NetChop 20S 3.0 In-vitro proteasome digestion model	0.100	0.065	0.241	0.596	0.519	0.854	0.909	0.857	0.102	0.846	0.298	0.932	0.876	0.163	0.949	0.060	0.255
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.357	0.060	0.904	0.021	0.936	0.042	0.245	0.412	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.229	0.247	0.421	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	N	M	N	N	N	M	N	M	N	N	N	N	N	M
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa H2 G3 : 0.057 log2 depletion	FAP [N2] / fap-dipeptidyl H2 G3 : N
XPNPEP1 [N1] / app1-xp K1 H2 : N	ENPEP [N1] / enpep-acidic K1 H2 : N
TPP2 [N3] / tpp2-tripeptidyl G3 K4 : M	ANPEP [N1] / anpep-ala K1 H2 : N
NPEPPS [N1] / npepps-n-terminal K1 H2 : M	DPP8 [N2] / dpp8-xp-xa H2 G3 : N
ACE [C2] / ace-dipeptidyl T16 G17 : M	DPP9 [N2] / dpp9-xp-xa H2 G3 : N
CPB2 (active assumption) [C1] / cpb2-basic G17 L18 : N	ERAP2 [N1] / erap2-basic K1 H2 : M
CPN1 [C1] / cpn-basic G17 L18 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp K1 H2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

EPG5 p.Ala869Thr

JLF V2 | 23 aa | bonds 1-22 | segment 1/1

KELPLYLWQPSTSEIAVIRDWKK

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
	K E	E L	L P	P L	L Y	Y L	L W	W Q	Q P	P S	S T	T S	S E	E I	I A	A V	V I	I R	R D	D W	W K	K K
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.226	0.017	0.014	0.006	0.024	0.036	0.084	0.902	0.221	0.176	0.092	0.073	0.046	0.132	0.562	0.129	0.434	0.436	0.425	0.327	0.973	0.754
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.030	0.007	0.004	0.013	0.024	0.029	0.222	0.761	0.120	0.128	0.076	0.048	0.062	0.086	0.610	0.150	0.325	0.520	0.565	0.301	0.945	0.813
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.639	0.026	0.799	0.024	0.927	0.642	0.853	0.920	0.035	0.430	0.030	0.059	0.067	0.208	0.549	0.047	0.735	0.561	0.656	0.056	0.920	0.253
NetChop 20S 3.0 In-vitro proteasome digestion model	0.160	0.426	0.933	0.060	0.060	0.768	0.951	0.693	0.219	0.403	0.032	0.101	0.424	0.425	0.780	0.559	0.170	0.642	0.940	0.134	0.252	0.079
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.758	0.014	0.540	0.164	0.362	0.149	0.116	0.751	0.094	0.208	0.445	0.246	0.031	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.326	0.519	0.216	0.351	0.296	0.169	0.513	0.374	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	M	N	N	N	N	N	M	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	M	N	M	M	M	N	N	N	N	N	N	N	M	N	N	M	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa | E2|L3 : 0.011 log2 depletion

XPNPEP1 [N1] / app1-xp | K1|E2 : N

TPP2 [N3] / tpp2-tripeptidyl | L3|P4 : N

NPEPPS [N1] / npepps-n-terminal | K1|E2 : M

ACE [C2] / ace-dipeptidyl | W21|K22 : M

CPB2 (active assumption) [C1] / cpb2-basic | K22|K23 : M

CPN1 [C1] / cpn-basic | K22|K23 : M

XPNPEP2 [N1] / app2-xp | K1|E2 : N

FAP [N2] / fap-dipeptidyl | E2|L3 : N

ENPEP [N1] / enpep-acidic | K1|E2 : N

ANPEP [N1] / anpep-ala | K1|E2 : N

DPP8 [N2] / dpp8-xp-xa | E2|L3 : N

DPP9 [N2] / dpp9-xp-xa | E2|L3 : N

ERAP2 [N1] / erap2-basic | K1|E2 : M

ERAP1 [N1] / eramer-step | no assessed bond : NA

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

EPG5 p.Ala869Thr

JLF V1 | 20 aa | bonds 1-19 | segment 1/1

KELPLYLWQPSTSEIAVIRD

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
	K E	E L	L P	P L	L Y	Y L	L W	W Q	Q P	P S	S T	T S	S E	E I	I A	A V	V I	I R	R D
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.226	0.017	0.014	0.006	0.024	0.036	0.084	0.902	0.221	0.176	0.092	0.073	0.093	0.205	0.661	0.209	0.592	0.832	0.849
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.030	0.007	0.004	0.013	0.024	0.029	0.222	0.761	0.120	0.128	0.076	0.048	0.065	0.132	0.698	0.249	0.516	0.739	0.868
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.639	0.026	0.799	0.024	0.927	0.642	0.853	0.920	0.035	0.430	0.030	0.059	0.147	0.264	0.651	0.124	0.896	0.724	0.781
NetChop 20S 3.0 In-vitro proteasome digestion model	0.160	0.426	0.933	0.060	0.060	0.768	0.951	0.693	0.219	0.403	0.032	0.101	0.424	0.425	0.780	0.559	0.170	0.893	0.854
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.758	0.014	0.540	0.164	0.362	0.149	0.116	0.751	0.094	0.208	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.326	0.519	0.216	0.351	0.296	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	M	N	N	N	N	N	M	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	M	N	M	M	M	N	N	N	N	N	N	N	M	N	N	M	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa E2 L3 : 0.011 log2 depletion	FAP [N2] / fap-dipeptidyl E2 L3 : N
XPNPEP1 [N1] / app1-xp K1 E2 : N	ENPEP [N1] / enpep-acidic K1 E2 : N
TPP2 [N3] / tpp2-tripeptidyl L3 P4 : N	ANPEP [N1] / anpep-ala K1 E2 : N
NPEPPS [N1] / npepps-n-terminal K1 E2 : M	DPP8 [N2] / dpp8-xp-xa E2 L3 : N
ACE [C2] / ace-dipeptidyl I18 R19 : N	DPP9 [N2] / dpp9-xp-xa E2 L3 : N
CPB2 (active assumption) [C1] / cpb2-basic R19 D20 : N	ERAP2 [N1] / erap2-basic K1 E2 : M
CPN1 [C1] / cpn-basic R19 D20 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp K1 E2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

EPG5 p.Ala869Thr

CeGaT | 23 aa | bonds 1-22 | segment 1/1

KKSVIRTLSTIDDVEDRENEKGR

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
	K K	K S	S V	V I	I R	R T	T L	L S	S T	T I	I D	D D	D V	V E	E D	D R	R E	E N	N E	E K	K G	G R
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.415	0.016	0.005	0.011	0.023	0.043	0.027	0.827	0.099	0.280	0.679	0.189	0.102	0.208	0.049	0.305	0.345	0.172	0.235	0.185	0.721	0.378
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.062	0.002	0.002	0.009	0.020	0.029	0.062	0.641	0.101	0.122	0.765	0.086	0.102	0.132	0.118	0.174	0.338	0.142	0.233	0.324	0.635	0.455
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.489	0.323	0.033	0.270	0.368	0.672	0.060	0.842	0.057	0.039	0.891	0.039	0.171	0.749	0.037	0.042	0.681	0.044	0.034	0.028	0.924	0.186
NetChop 20S 3.0 In-vitro proteasome digestion model	0.056	0.230	0.352	0.422	0.881	0.967	0.115	0.984	0.457	0.349	0.448	0.583	0.291	0.538	0.199	0.709	0.758	0.121	0.045	0.870	0.046	0.190
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.946	0.028	0.073	0.884	0.074	0.184	0.675	0.119	0.218	0.624	0.103	0.345	0.098	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.482	0.233	0.296	0.260	0.468	0.356	0.415	0.272	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	M	N	N	M	N	N	M	N	N	N	N	N	N	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa no assessed bond : NA	FAP [N2] / fap-dipeptidyl K2 S3 : N
XPNPEP1 [N1] / app1-xp K1 K2 : N	ENPEP [N1] / enpep-acidic K1 K2 : N
TPP2 [N3] / tpp2-tripeptidyl S3 V4 : M	ANPEP [N1] / anpep-ala K1 K2 : N
NPEPPS [N1] / npepps-n-terminal K1 K2 : M	DPP8 [N2] / dpp8-xp-xa K2 S3 : N
ACE [C2] / ace-dipeptidyl K21 G22 : M	DPP9 [N2] / dpp9-xp-xa K2 S3 : N
CPB2 (active assumption) [C1] / cpb2-basic G22 R23 : M	ERAP2 [N1] / erap2-basic K1 K2 : M
CPN1 [C1] / cpn-basic G22 R23 : M	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp K1 K2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

EXOC4 p.Ser34Ile

JLF V3;JLF V2 | 23 aa | bonds 1-22 | segment 1/1

KKSVIRTLSTIDDVEDRENEKGR

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
	K K	K S	S V	V I	I R	R T	T L	L S	S T	T I	I D	D D	D V	V E	E D	D R	R E	E N	N E	E K	K G	G R
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.415	0.016	0.005	0.011	0.023	0.043	0.027	0.827	0.099	0.280	0.679	0.189	0.102	0.208	0.049	0.305	0.345	0.172	0.235	0.185	0.721	0.378
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.062	0.002	0.002	0.009	0.020	0.029	0.062	0.641	0.101	0.122	0.765	0.086	0.102	0.132	0.118	0.174	0.338	0.142	0.233	0.324	0.635	0.455
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.489	0.323	0.033	0.270	0.368	0.672	0.060	0.842	0.057	0.039	0.891	0.039	0.171	0.749	0.037	0.042	0.681	0.044	0.034	0.028	0.924	0.186
NetChop 20S 3.0 In-vitro proteasome digestion model	0.056	0.230	0.352	0.422	0.881	0.967	0.115	0.984	0.457	0.349	0.448	0.583	0.291	0.538	0.199	0.709	0.758	0.121	0.045	0.870	0.046	0.190
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.946	0.028	0.073	0.884	0.074	0.184	0.675	0.119	0.218	0.624	0.103	0.345	0.098	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.482	0.233	0.296	0.260	0.468	0.356	0.415	0.272	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	M	N	N	M	N	N	M	N	N	N	N	N	N	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa no assessed bond : NA	FAP [N2] / fap-dipeptidyl K2 S3 : N
XPNPEP1 [N1] / app1-xp K1 K2 : N	ENPEP [N1] / enpep-acidic K1 K2 : N
TPP2 [N3] / tpp2-tripeptidyl S3 V4 : M	ANPEP [N1] / anpep-ala K1 K2 : N
NPEPPS [N1] / npepps-n-terminal K1 K2 : M	DPP8 [N2] / dpp8-xp-xa K2 S3 : N
ACE [C2] / ace-dipeptidyl K21 G22 : M	DPP9 [N2] / dpp9-xp-xa K2 S3 : N
CPB2 (active assumption) [C1] / cpb2-basic G22 R23 : M	ERAP2 [N1] / erap2-basic K1 K2 : M
CPN1 [C1] / cpn-basic G22 R23 : M	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp K1 K2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

EXOC4 p.Ser34Ile

CeGaT | 17 aa | bonds 1-16 | segment 1/1

SVIRTLSTIDDVEDREN

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	S V	V I	I R	R T	T L	L S	S T	T I	I D	D D	D V	V E	E D	D R	R E	E N
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.010	0.011	0.026	0.032	0.011	0.210	0.031	0.286	0.679	0.213	0.195	0.167	0.146	0.327	0.552	0.242
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.002	0.003	0.019	0.018	0.011	0.028	0.101	0.168	0.765	0.112	0.139	0.146	0.158	0.247	0.483	0.453
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.030	0.486	0.533	0.854	0.058	0.844	0.054	0.032	0.891	0.044	0.169	0.603	0.042	0.087	0.738	0.131
NetChop 20S 3.0 In-vitro proteasome digestion model	0.167	0.275	0.805	0.967	0.115	0.984	0.457	0.349	0.448	0.583	0.291	0.538	0.199	0.709	0.755	0.380
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.073	0.884	0.074	0.184	0.675	0.119	0.218	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.296	0.260	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	M	N	N	M	N	N	M	N	N	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa V2 I3 : -0.015 log2 depletion	FAP [N2] / fap-dipeptidyl V2 I3 : N
XPNPEP1 [N1] / app1-xp S1 V2 : N	ENPEP [N1] / enpep-acidic S1 V2 : N
TPP2 [N3] / tpp2-tripeptidyl I3 R4 : M	ANPEP [N1] / anpep-ala S1 V2 : N
NPEPPS [N1] / npepps-n-terminal S1 V2 : M	DPP8 [N2] / dpp8-xp-xa V2 I3 : N
ACE [C2] / ace-dipeptidyl R15 E16 : M	DPP9 [N2] / dpp9-xp-xa V2 I3 : N
CPB2 (active assumption) [C1] / cpb2-basic E16 N17 : N	ERAP2 [N1] / erap2-basic S1 V2 : N
CPN1 [C1] / cpn-basic E16 N17 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp S1 V2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

GLIS3 p.Ser775fs

JLF V3 | 80 aa | bonds 1-24 | segment 1/4

QPPESSSSFFNTAKNTASLYPAAIR

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
	Q P	P P	P E	E S	S S	S S	S S	S F	F F	F N	N T	T A	A K	K N	N T	T A	A S	S L	L Y	Y P	P A	A A	A I	I R
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.016	0.003	0.006	0.014	0.006	0.010	0.019	0.266	0.869	0.699	0.214	0.159	0.459	0.633	0.065	0.177	0.570	0.189	0.702	0.658	0.060	0.143	0.280	0.458
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.003	0.003	0.013	0.034	0.011	0.028	0.033	0.218	0.797	0.637	0.101	0.069	0.497	0.519	0.116	0.170	0.443	0.131	0.582	0.602	0.061	0.124	0.225	0.421
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.032	0.045	0.092	0.036	0.056	0.033	0.051	0.045	0.968	0.924	0.026	0.039	0.893	0.959	0.330	0.061	0.373	0.024	0.902	0.933	0.030	0.235	0.039	0.871
NetChop 20S 3.0 In-vitro proteasome digestion model	0.283	0.063	0.025	0.031	0.174	0.369	0.468	0.334	0.776	0.833	0.185	0.163	0.864	0.139	0.201	0.648	0.912	0.154	0.625	0.391	0.198	0.103	0.191	0.961
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.038	0.611	0.427	0.022	0.243	0.662	0.535	0.050	0.137	0.747	0.017	0.560	0.456	0.050	0.451	0.136	0.475
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.357	0.316	0.395	0.304	0.315	0.348	0.359	0.157	0.205	0.478	0.310	0.187

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
MME / mme-hydrophobic (preferred)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa P2 P3 : 0.113 log2 depletion	FAP [N2] / fap-dipeptidyl P2 P3 : N
XPNPEP1 [N1] / app1-xp Q1 P2 : M	ENPEP [N1] / enpep-acidic Q1 P2 : N
TPP2 [N3] / tpp2-tripeptidyl P3 E4 : N	ANPEP [N1] / anpep-ala Q1 P2 : N
NPEPPS [N1] / npepps-n-terminal Q1 P2 : N	DPP8 [N2] / dpp8-xp-xa P2 P3 : N
ACE [C2] / ace-dipeptidyl A78 F79 : M	DPP9 [N2] / dpp9-xp-xa P2 P3 : N
CPB2 (active assumption) [C1] / cpb2-basic F79 V80 : N	ERAP2 [N1] / erap2-basic Q1 P2 : N
CPN1 [C1] / cpn-basic F79 V80 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp Q1 P2 : M	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

GLIS3 p.Ser775fs

JLF V3 | 80 aa | bonds 25-48 | segment 2/4

RFTPEVLSARNKLFFSTKWYPCPWI

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
	R F	F T	T P	P E	E V	V L	L S	S A	A R	R N	N K	K L	L F	F F	F S	S T	T K	K W	W Y	Y P	P C	C P	P W	W I
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.497	0.926	0.141	0.238	0.111	0.636	0.752	0.103	0.740	0.785	0.073	0.276	0.358	0.360	0.570	0.047	0.420	0.553	0.636	0.356	0.050	0.090	0.058	0.325
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.396	0.895	0.104	0.255	0.126	0.607	0.774	0.086	0.771	0.673	0.099	0.282	0.329	0.368	0.636	0.054	0.447	0.468	0.564	0.352	0.041	0.057	0.050	0.370
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.193	0.957	0.028	0.136	0.034	0.845	0.820	0.275	0.699	0.358	0.041	0.088	0.941	0.619	0.264	0.024	0.346	0.783	0.902	0.893	0.028	0.023	0.024	0.853
NetChop 20S 3.0 In-vitro proteasome digestion model	0.689	0.887	0.251	0.323	0.050	0.141	0.945	0.302	0.932	0.332	0.348	0.711	0.800	0.402	0.920	0.049	0.958	0.258	0.775	0.827	0.051	0.277	0.147	0.055
NetCleave-I HLA MHC-I C-terminal processing proxy	0.111	0.806	0.060	0.301	0.153	0.188	0.787	0.022	0.475	0.534	0.050	0.476	0.390	0.140	0.737	0.024	0.351	0.260	0.686	0.073	0.155	0.154	0.094	0.508
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	0.414	0.328	0.206	0.173	0.431	0.159	0.513	0.352	0.281	0.323	0.380	0.389	0.422	0.294	0.327	0.191	0.342	0.471	0.232	0.145	0.439	0.116	0.260	0.323

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
MME / mme-hydrophobic (preferred)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa P2 P3 : 0.113 log2 depletion	FAP [N2] / fap-dipeptidyl P2 P3 : N
XPNPEP1 [N1] / app1-xp Q1 P2 : M	ENPEP [N1] / enpep-acidic Q1 P2 : N
TPP2 [N3] / tpp2-tripeptidyl P3 E4 : N	ANPEP [N1] / anpep-ala Q1 P2 : N
NPEPPS [N1] / npepps-n-terminal Q1 P2 : N	DPP8 [N2] / dpp8-xp-xa P2 P3 : N
ACE [C2] / ace-dipeptidyl A78 F79 : M	DPP9 [N2] / dpp9-xp-xa P2 P3 : N
CPB2 (active assumption) [C1] / cpb2-basic F79 V80 : N	ERAP2 [N1] / erap2-basic Q1 P2 : N
CPN1 [C1] / cpn-basic F79 V80 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp Q1 P2 : M	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

GLIS3 p.Ser775fs

JLF V3 | 80 aa | bonds 49-72 | segment 3/4

ILWAAAEVLSPTLPRFPENCAACQL

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72
	I L	L W	W A	A A	A A	A E	E V	V L	L S	S P	P T	T L	L P	P R	R F	F P	P E	E N	N C	C A	A A	A C	C Q	Q L
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.443	0.709	0.920	0.720	0.303	0.208	0.040	0.238	0.665	0.066	0.137	0.092	0.529	0.298	0.541	0.868	0.113	0.209	0.217	0.189	0.609	0.647	0.534	0.184
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.388	0.717	0.862	0.612	0.299	0.239	0.029	0.303	0.663	0.047	0.154	0.139	0.633	0.227	0.636	0.810	0.114	0.168	0.202	0.344	0.637	0.622	0.537	0.124
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.097	0.719	0.965	0.844	0.276	0.846	0.029	0.534	0.973	0.024	0.029	0.048	0.923	0.088	0.629	0.863	0.025	0.134	0.199	0.067	0.898	0.831	0.097	0.091
NetChop 20S 3.0 In-vitro proteasome digestion model	0.208	0.959	0.925	0.688	0.837	0.757	0.325	0.151	0.680	0.498	0.166	0.028	0.740	0.366	0.121	0.462	0.385	0.068	0.076	0.851	0.533	0.538	0.490	0.465
NetCleave-I HLA MHC-I C-terminal processing proxy	0.177	0.311	0.647	0.048	0.571	0.332	0.128	0.210	0.815	0.027	0.627	0.036	0.834	0.051	0.301	0.674	0.093	0.125	0.545	0.233	0.404	0.350	0.229	0.069
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	0.155	0.560	0.239	0.284	0.386	0.246	0.273	0.239	0.399	0.221	0.189	0.657	0.192	0.233	0.573	0.147	0.287	0.290	0.544	0.086	0.321	0.512	0.087	0.326

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
MME / mme-hydrophobic (preferred)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa | P2|P3 : 0.113 log2 depletion

XPNPEP1 [N1] / app1-xp | Q1|P2 : M

TPP2 [N3] / tpp2-tripeptidyl | P3|E4 : N

NPEPPS [N1] / npepps-n-terminal | Q1|P2 : N

ACE [C2] / ace-dipeptidyl | A78|F79 : M

CPB2 (active assumption) [C1] / cpb2-basic | F79|V80 : N

CPN1 [C1] / cpn-basic | F79|V80 : N

XPNPEP2 [N1] / app2-xp | Q1|P2 : M

FAP [N2] / fap-dipeptidyl | P2|P3 : N

ENPEP [N1] / enpep-acidic | Q1|P2 : N

ANPEP [N1] / anpep-ala | Q1|P2 : N

DPP8 [N2] / dpp8-xp-xa | P2|P3 : N

DPP9 [N2] / dpp9-xp-xa | P2|P3 : N

ERAP2 [N1] / erap2-basic | Q1|P2 : N

ERAP1 [N1] / eramer-step | no assessed bond : NA

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

GLIS3 p.Ser775fs

JLF V3 | 80 aa | bonds 73-79 | segment 4/4

LLQCGAFV

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	73	74	75	76	77	78	79
	L L	L Q	Q C	C G	G A	A F	F V
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.624	0.586	0.192	0.322	0.203	0.305	0.659
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.534	0.724	0.188	0.275	0.165	0.293	0.680
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.656	0.332	0.029	0.025	0.038	0.098	0.379
NetChop 20S 3.0 In-vitro proteasome digestion model	0.827	0.660	0.641	0.701	0.801	0.321	0.188
NetCleave-I HLA MHC-I C-terminal processing proxy	0.600	0.516	0.071	0.398	0.123	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	0.391	0.198	0.450	0.093	0.458	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	NA	NA	NA	NA	NA	NA	NA
MME / mme-hydrophobic (preferred)	NA	NA	NA	NA	NA	NA	NA
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa | P2|P3 : 0.113 log2 depletion

XPNPEP1 [N1] / app1-xp | Q1|P2 : M

TPP2 [N3] / tpp2-tripeptidyl | P3|E4 : N

NPEPPS [N1] / npepps-n-terminal | Q1|P2 : N

ACE [C2] / ace-dipeptidyl | A78|F79 : M

CPB2 (active assumption) [C1] / cpb2-basic | F79|V80 : N

CPN1 [C1] / cpn-basic | F79|V80 : N

XPNPEP2 [N1] / app2-xp | Q1|P2 : M

FAP [N2] / fap-dipeptidyl | P2|P3 : N

ENPEP [N1] / enpep-acidic | Q1|P2 : N

ANPEP [N1] / anpep-ala | Q1|P2 : N

DPP8 [N2] / dpp8-xp-xa | P2|P3 : N

DPP9 [N2] / dpp9-xp-xa | P2|P3 : N

ERAP2 [N1] / erap2-basic | Q1|P2 : N

ERAP1 [N1] / eramer-step | no assessed bond : NA

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

H1-2 p.Ala197_Lys201del

CeGaT | 16 aa | bonds 1-15 | segment 1/1

AAKAVKPKVVKPKKAA

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	A A	A K	K A	A V	V K	K P	P K	K V	V V	V K	K P	P K	K K	K A	A A
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.476	0.021	0.020	0.008	0.019	0.022	0.008	0.525	0.360	0.523	0.725	0.531	0.832	0.753	0.674
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.001	0.001	0.009	0.004	0.007	0.029	0.011	0.316	0.311	0.425	0.642	0.360	0.769	0.633	0.538
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.262	0.231	0.481	0.204	0.236	0.076	0.029	0.183	0.695	0.273	0.462	0.422	0.889	0.973	0.448
NetChop 20S 3.0 In-vitro proteasome digestion model	0.082	0.315	0.066	0.047	0.362	0.036	0.157	0.028	0.041	0.713	0.045	0.039	0.028	0.066	0.921
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.371	0.135	0.501	0.240	0.151	0.574	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.396	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	M	N	N	N	N	M	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa A2 K3 : 3.426 log2 depletion	FAP [N2] / fap-dipeptidyl A2 K3 : N
XPNPEP1 [N1] / app1-xp A1 A2 : N	ENPEP [N1] / enpep-acidic A1 A2 : N
TPP2 [N3] / tpp2-tripeptidyl K3 A4 : M	ANPEP [N1] / anpep-ala A1 A2 : M
NPEPPS [N1] / npepps-n-terminal A1 A2 : M	DPP8 [N2] / dpp8-xp-xa A2 K3 : M
ACE [C2] / ace-dipeptidyl K14 A15 : M	DPP9 [N2] / dpp9-xp-xa A2 K3 : M
CPB2 (active assumption) [C1] / cpb2-basic A15 A16 : N	ERAP2 [N1] / erap2-basic A1 A2 : N
CPN1 [C1] / cpn-basic A15 A16 : N	ERAP1 [N1] / eramer-step A1 A2 : 0.095 PWM specificity
XPNPEP2 [N1] / app2-xp A1 A2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

KDM3B p.Pro633Ser

JLF V2;JLF V1 | 26 aa | bonds 1-24 | segment 1/2

HENLFLQPPKLSREESSNPFLAFVE

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
	H E	E N	N L	L F	F L	L Q	Q P	P P	P K	K L	L S	S R	R E	E E	E S	S S	S N	N P	P F	F L	L A	A F	F V	V E
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.020	0.014	0.006	0.008	0.015	0.016	0.016	0.177	0.329	0.523	0.874	0.454	0.786	0.077	0.122	0.034	0.031	0.048	0.087	0.354	0.604	0.416	0.863	0.744
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.010	0.013	0.004	0.009	0.018	0.015	0.019	0.127	0.251	0.598	0.877	0.361	0.729	0.076	0.107	0.057	0.034	0.044	0.073	0.344	0.505	0.520	0.759	0.667
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.148	0.050	0.083	0.931	0.640	0.602	0.043	0.051	0.286	0.690	0.978	0.038	0.972	0.123	0.084	0.055	0.024	0.024	0.028	0.907	0.644	0.304	0.937	0.924
NetChop 20S 3.0 In-vitro proteasome digestion model	0.520	0.547	0.526	0.496	0.234	0.613	0.207	0.085	0.027	0.026	0.796	0.582	0.767	0.175	0.625	0.508	0.109	0.364	0.063	0.044	0.932	0.691	0.451	0.110
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.453	0.279	0.198	0.665	0.027	0.738	0.048	0.396	0.239	0.214	0.224	0.138	0.279	0.521	0.107	0.749	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.372	0.262	0.392	0.348	0.257	0.216	0.286	0.531	0.227	0.223	0.655	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	M	M	N	N	N	N	N	N	N	N	N	M	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	M	M	M	N	N	N	N	M	N	N	N	N	N	N	N	M	M	N	M	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa E2 N3 : 0.028 log2 depletion	FAP [N2] / fap-dipeptidyl E2 N3 : N
XPNPEP1 [N1] / app1-xp H1 E2 : N	ENPEP [N1] / enpep-acidic H1 E2 : N
TPP2 [N3] / tpp2-tripeptidyl N3 L4 : M	ANPEP [N1] / anpep-ala H1 E2 : N
NPEPPS [N1] / npepps-n-terminal H1 E2 : M	DPP8 [N2] / dpp8-xp-xa E2 N3 : N
ACE [C2] / ace-dipeptidyl V24 E25 : M	DPP9 [N2] / dpp9-xp-xa E2 N3 : N
CPB2 (active assumption) [C1] / cpb2-basic E25 K26 : M	ERAP2 [N1] / erap2-basic H1 E2 : N
CPN1 [C1] / cpn-basic E25 K26 : M	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp H1 E2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

KDM3B p.Pro633Ser

JLF V2;JLF V1 | 26 aa | bonds 25-25 | segment 2/2

EK

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	25
	E K
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.424
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.443
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.120
NetChop 20S 3.0 In-vitro proteasome digestion model	0.810
NetCleave-I HLA MHC-I C-terminal processing proxy	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N
MME / mme-hydrophobic (preferred)	N
FAP / fap-endo-gp (required)	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa E2 N3 : 0.028 log2 depletion	FAP [N2] / fap-dipeptidyl E2 N3 : N
XPNPEP1 [N1] / app1-xp H1 E2 : N	ENPEP [N1] / enpep-acidic H1 E2 : N
TPP2 [N3] / tpp2-tripeptidyl N3 L4 : M	ANPEP [N1] / anpep-ala H1 E2 : N
NPEPPS [N1] / npepps-n-terminal H1 E2 : M	DPP8 [N2] / dpp8-xp-xa E2 N3 : N
ACE [C2] / ace-dipeptidyl V24 E25 : M	DPP9 [N2] / dpp9-xp-xa E2 N3 : N
CPB2 (active assumption) [C1] / cpb2-basic E25 K26 : M	ERAP2 [N1] / erap2-basic H1 E2 : N
CPN1 [C1] / cpn-basic E25 K26 : M	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp H1 E2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

KDM3B p.Pro633Ser

CeGaT | 17 aa | bonds 1-16 | segment 1/1

PPKLSREESSNPFLAFV

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	P P	P K	K L	L S	S R	R E	E E	E S	S S	S N	N P	P F	F L	L A	A F	F V
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.005	0.013	0.038	0.042	0.018	0.047	0.066	0.162	0.034	0.062	0.084	0.085	0.355	0.689	0.625	0.955
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.004	0.004	0.023	0.019	0.020	0.031	0.077	0.103	0.057	0.035	0.072	0.103	0.374	0.643	0.627	0.910
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.036	0.191	0.847	0.974	0.043	0.953	0.118	0.082	0.055	0.024	0.024	0.029	0.929	0.768	0.410	0.937
NetChop 20S 3.0 In-vitro proteasome digestion model	0.033	0.028	0.025	0.796	0.582	0.767	0.175	0.625	0.508	0.109	0.364	0.063	0.044	0.932	0.682	0.790
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.396	0.239	0.214	0.224	0.138	0.279	0.521	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.531	0.227	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	M	N	N	N	N	N	N	N	N	N	M	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	M	N	N	N	N	N	N	N	N	M	M	N	M	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa no assessed bond : NA	FAP [N2] / fap-dipeptidyl P2 K3 : M
XPNPEP1 [N1] / app1-xp P1 P2 : M	ENPEP [N1] / enpep-acidic P1 P2 : N
TPP2 [N3] / tpp2-tripeptidyl K3 L4 : M	ANPEP [N1] / anpep-ala P1 P2 : N
NPEPPS [N1] / npepps-n-terminal P1 P2 : N	DPP8 [N2] / dpp8-xp-xa P2 K3 : M
ACE [C2] / ace-dipeptidyl A15 F16 : M	DPP9 [N2] / dpp9-xp-xa P2 K3 : M
CPB2 (active assumption) [C1] / cpb2-basic F16 V17 : N	ERAP2 [N1] / erap2-basic P1 P2 : N
CPN1 [C1] / cpn-basic F16 V17 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp P1 P2 : M	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

KIF1C p.Glu456Lys

JLF V2;JLF V1 | 21 aa | bonds 1-20 | segment 1/1

TEKIIAELNKTWEEKLRKTEA

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
	T E	E K	K I	I I	I A	A E	E L	L N	N K	K T	T W	W E	E E	E K	K L	L R	R K	K T	T E	E A
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.009	0.014	0.013	0.011	0.018	0.043	0.065	0.819	0.066	0.344	0.141	0.612	0.091	0.164	0.604	0.677	0.480	0.820	0.400	0.541
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.002	0.006	0.007	0.019	0.018	0.045	0.044	0.622	0.081	0.231	0.110	0.677	0.093	0.120	0.543	0.575	0.490	0.811	0.391	0.597
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.025	0.025	0.640	0.411	0.117	0.147	0.031	0.970	0.028	0.104	0.032	0.952	0.024	0.149	0.830	0.733	0.147	0.914	0.165	0.039
NetChop 20S 3.0 In-vitro proteasome digestion model	0.048	0.773	0.117	0.230	0.815	0.778	0.789	0.341	0.078	0.762	0.095	0.646	0.104	0.959	0.090	0.182	0.867	0.137	0.178	0.886
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.954	0.031	0.698	0.042	0.910	0.012	0.225	0.120	0.753	0.154	0.164	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.279	0.330	0.468	0.277	0.351	0.315	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	M	M	N	N	M	N	N	N	N	N	N	N	M	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa E2 K3 : 0.000 log2 depletion	FAP [N2] / fap-dipeptidyl E2 K3 : N
XPNPEP1 [N1] / app1-xp T1 E2 : N	ENPEP [N1] / enpep-acidic T1 E2 : N
TPP2 [N3] / tpp2-tripeptidyl K3 I4 : M	ANPEP [N1] / anpep-ala T1 E2 : N
NPEPPS [N1] / npepps-n-terminal T1 E2 : M	DPP8 [N2] / dpp8-xp-xa E2 K3 : N
ACE [C2] / ace-dipeptidyl T19 E20 : M	DPP9 [N2] / dpp9-xp-xa E2 K3 : N
CPB2 (active assumption) [C1] / cpb2-basic E20 A21 : N	ERAP2 [N1] / erap2-basic T1 E2 : N
CPN1 [C1] / cpn-basic E20 A21 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp T1 E2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

MAP2 p.Leu867fs

JLF V3 | 23 aa | bonds 1-22 | segment 1/1

KTVRIYQGRVVPFTKALMIKFEE

Source minimal epitope: RVVPFTKAL (SLP 9-17)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
	K T	T V	V R	R I	I Y	Y Q	Q G	G R	R V	V V	V P	P F	F T	T K	K A	A L	L M	M I	I K	K F	F E	E E
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.235	0.011	0.017	0.011	0.019	0.061	0.031	0.087	0.416	0.072	0.304	0.160	0.730	0.063	0.527	0.571	0.864	0.684	0.847	0.599	0.896	0.210
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.002	0.003	0.005	0.015	0.015	0.034	0.040	0.069	0.481	0.057	0.240	0.276	0.585	0.056	0.433	0.507	0.860	0.594	0.821	0.535	0.863	0.327
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.259	0.027	0.791	0.209	0.579	0.677	0.041	0.070	0.056	0.178	0.230	0.036	0.729	0.025	0.548	0.836	0.964	0.565	0.967	0.772	0.859	0.033
NetChop 20S 3.0 In-vitro proteasome digestion model	0.217	0.118	0.964	0.823	0.906	0.923	0.628	0.832	0.604	0.145	0.945	0.201	0.040	0.182	0.880	0.721	0.935	0.086	0.965	0.140	0.848	0.803
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.115	0.536	0.060	0.628	0.085	0.591	0.058	0.707	0.200	0.685	0.143	0.567	0.082	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.368	0.212	0.350	0.368	0.329	0.333	0.120	0.533	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	M	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	M	M	N	N	N	N	N	N	M	N	N	N	M	N	M	N	M	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa T2 V3 : 0.486 log2 depletion	FAP [N2] / fap-dipeptidyl T2 V3 : N
XPNPEP1 [N1] / app1-xp K1 T2 : N	ENPEP [N1] / enpep-acidic K1 T2 : N
TPP2 [N3] / tpp2-tripeptidyl V3 R4 : M	ANPEP [N1] / anpep-ala K1 T2 : N
NPEPPS [N1] / npepps-n-terminal K1 T2 : M	DPP8 [N2] / dpp8-xp-xa T2 V3 : N
ACE [C2] / ace-dipeptidyl F21 E22 : N	DPP9 [N2] / dpp9-xp-xa T2 V3 : N
CPB2 (active assumption) [C1] / cpb2-basic E22 E23 : N	ERAP2 [N1] / erap2-basic K1 T2 : M
CPN1 [C1] / cpn-basic E22 E23 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp K1 T2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

MAP2 p.Leu867fs

JLF V2;JLF V1 | 24 aa | bonds 1-23 | segment 1/1

KTVRIYQGRVVPFTKALMIKFEEI

Source minimal epitope: RVVPFTKAL (SLP 9-17)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
	K T	T V	V R	R I	I Y	Y Q	Q G	G R	R V	V V	V P	P F	F T	T K	K A	A L	L M	M I	I K	K F	F E	E E	E I
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.235	0.011	0.017	0.011	0.019	0.061	0.031	0.087	0.416	0.072	0.304	0.160	0.730	0.063	0.527	0.548	0.871	0.691	0.823	0.464	0.808	0.134	0.272
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.002	0.003	0.005	0.015	0.015	0.034	0.040	0.069	0.481	0.057	0.240	0.276	0.585	0.056	0.433	0.423	0.863	0.584	0.812	0.455	0.779	0.142	0.398
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.259	0.027	0.791	0.209	0.579	0.677	0.041	0.070	0.056	0.178	0.230	0.036	0.729	0.025	0.548	0.823	0.960	0.448	0.938	0.590	0.487	0.029	0.038
NetChop 20S 3.0 In-vitro proteasome digestion model	0.217	0.118	0.964	0.823	0.906	0.923	0.628	0.832	0.604	0.145	0.945	0.201	0.040	0.182	0.880	0.721	0.935	0.086	0.965	0.140	0.921	0.902	0.485
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.115	0.536	0.060	0.628	0.085	0.591	0.058	0.707	0.200	0.685	0.143	0.567	0.082	0.834	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.368	0.212	0.350	0.368	0.329	0.333	0.120	0.533	0.427	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	M	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	M	M	N	N	N	N	N	N	M	N	N	N	M	N	M	N	M	N	N	M
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa | T2|V3 : 0.486 log2 depletion

XPNPEP1 [N1] / app1-xp | K1|T2 : N

TPP2 [N3] / tpp2-tripeptidyl | V3|R4 : M

NPEPPS [N1] / npepps-n-terminal | K1|T2 : M

ACE [C2] / ace-dipeptidyl | E22|E23 : M

CPB2 (active assumption) [C1] / cpb2-basic | E23|I24 : N

CPN1 [C1] / cpn-basic | E23|I24 : N

XPNPEP2 [N1] / app2-xp | K1|T2 : N

FAP [N2] / fap-dipeptidyl | T2|V3 : N

ENPEP [N1] / enpep-acidic | K1|T2 : N

ANPEP [N1] / anpep-ala | K1|T2 : N

DPP8 [N2] / dpp8-xp-xa | T2|V3 : N

DPP9 [N2] / dpp9-xp-xa | T2|V3 : N

ERAP2 [N1] / erap2-basic | K1|T2 : M

ERAP1 [N1] / eramer-step | no assessed bond : NA

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

MT-ND5 p.Ala220Thr

CeGaT | 17 aa | bonds 1-16 | segment 1/1

PLLGLLLATAGKSAQLG

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	P L	L L	L G	G L	L L	L L	L A	A T	T A	A G	G K	K S	S A	A Q	Q L	L G
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.009	0.012	0.016	0.011	0.040	0.053	0.196	0.536	0.285	0.432	0.143	0.567	0.149	0.389	0.474	0.946
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.004	0.004	0.011	0.009	0.020	0.029	0.211	0.359	0.218	0.379	0.172	0.449	0.139	0.339	0.444	0.907
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.031	0.705	0.801	0.038	0.933	0.805	0.935	0.811	0.094	0.793	0.103	0.963	0.101	0.752	0.184	0.971
NetChop 20S 3.0 In-vitro proteasome digestion model	0.063	0.140	0.596	0.743	0.806	0.241	0.952	0.755	0.290	0.074	0.062	0.086	0.244	0.851	0.222	0.680
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.192	0.260	0.716	0.048	0.900	0.032	0.819	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.404	0.252	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	M	M	N	M	M	M	N	N	N	N	N	N	N	N	M	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa L2 L3 : 0.026 log2 depletion	FAP [N2] / fap-dipeptidyl L2 L3 : N
XPNPEP1 [N1] / app1-xp P1 L2 : N	ENPEP [N1] / enpep-acidic P1 L2 : N
TPP2 [N3] / tpp2-tripeptidyl L3 G4 : M	ANPEP [N1] / anpep-ala P1 L2 : N
NPEPPS [N1] / npepps-n-terminal P1 L2 : N	DPP8 [N2] / dpp8-xp-xa L2 L3 : N
ACE [C2] / ace-dipeptidyl Q15 L16 : M	DPP9 [N2] / dpp9-xp-xa L2 L3 : N
CPB2 (active assumption) [C1] / cpb2-basic L16 G17 : N	ERAP2 [N1] / erap2-basic P1 L2 : N
CPN1 [C1] / cpn-basic L16 G17 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp P1 L2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

MYO15B fs

JLF V2;JLF V1 | 17 aa | bonds 1-16 | segment 1/1

AGRRAQAPTRVLGLAPP

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	A G	G R	R R	R A	A Q	Q A	A P	P T	T R	R V	V L	L G	G L	L A	A P	P P
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.203	0.008	0.031	0.018	0.004	0.006	0.021	0.053	0.054	0.336	0.528	0.843	0.420	0.869	0.367	0.297
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.198	0.004	0.028	0.005	0.005	0.006	0.044	0.036	0.058	0.340	0.539	0.749	0.370	0.814	0.322	0.313
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.409	0.073	0.405	0.379	0.181	0.028	0.050	0.033	0.052	0.836	0.947	0.955	0.193	0.958	0.688	0.046
NetChop 20S 3.0 In-vitro proteasome digestion model	0.247	0.104	0.598	0.654	0.526	0.565	0.836	0.047	0.040	0.720	0.935	0.869	0.224	0.778	0.110	0.089
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.184	0.143	0.337	0.178	0.667	0.023	0.850	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.377	0.221	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	M	N	N	N	N	N	N	N	M
MME / mme-hydrophobic (preferred)	N	N	N	N	N	N	N	N	N	N	M	N	M	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa | no assessed bond : NA

XPNPEP1 [N1] / app1-xp | A1|G2 : N

TPP2 [N3] / tpp2-tripeptidyl | R3|R4 : M

NPEPPS [N1] / npepps-n-terminal | A1|G2 : N

ACE [C2] / ace-dipeptidyl | A15|P16 : N

CPB2 (active assumption) [C1] / cpb2-basic | P16|P17 : N

CPN1 [C1] / cpn-basic | P16|P17 : N

XPNPEP2 [N1] / app2-xp | A1|G2 : N

FAP [N2] / fap-dipeptidyl | G2|R3 : N

ENPEP [N1] / enpep-acidic | A1|G2 : N

ANPEP [N1] / anpep-ala | A1|G2 : M

DPP8 [N2] / dpp8-xp-xa | G2|R3 : N

DPP9 [N2] / dpp9-xp-xa | G2|R3 : N

ERAP2 [N1] / erap2-basic | A1|G2 : N

ERAP1 [N1] / eramer-step | no assessed bond : NA

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

MYO9A p.Asp523Val

JLF V2 | 25 aa | bonds 1-24 | segment 1/1

KKFDWIVFRINHALLNSKVLEHNTK

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
	K K	K F	F D	D W	W I	I V	V F	F R	R I	I N	N H	H A	A L	L L	L N	N S	S K	K V	V L	L E	E H	H N	N T	T K
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.252	0.005	0.009	0.007	0.059	0.019	0.016	0.775	0.857	0.811	0.074	0.147	0.220	0.753	0.502	0.159	0.131	0.737	0.520	0.584	0.193	0.780	0.337	0.338
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.104	0.002	0.004	0.029	0.036	0.009	0.065	0.723	0.880	0.722	0.069	0.098	0.223	0.724	0.451	0.113	0.170	0.586	0.485	0.566	0.157	0.661	0.292	0.406
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.484	0.207	0.104	0.023	0.192	0.067	0.183	0.590	0.341	0.939	0.058	0.030	0.334	0.956	0.937	0.043	0.024	0.574	0.349	0.943	0.060	0.581	0.084	0.138
NetChop 20S 3.0 In-vitro proteasome digestion model	0.248	0.067	0.959	0.889	0.599	0.590	0.583	0.981	0.975	0.940	0.485	0.963	0.417	0.733	0.157	0.083	0.699	0.107	0.230	0.968	0.846	0.616	0.040	0.176
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.743	0.228	0.587	0.058	0.359	0.284	0.565	0.271	0.067	0.066	0.717	0.284	0.439	0.045	0.704	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.271	0.365	0.285	0.342	0.251	0.433	0.264	0.306	0.268	0.400	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	M	N	N	M	N	M	N	M	N	N	N	M	M	N	N	N	N	M	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa no assessed bond : NA	FAP [N2] / fap-dipeptidyl K2 F3 : N
XPNPEP1 [N1] / app1-xp K1 K2 : N	ENPEP [N1] / enpep-acidic K1 K2 : N
TPP2 [N3] / tpp2-tripeptidyl F3 D4 : M	ANPEP [N1] / anpep-ala K1 K2 : N
NPEPPS [N1] / npepps-n-terminal K1 K2 : M	DPP8 [N2] / dpp8-xp-xa K2 F3 : N
ACE [C2] / ace-dipeptidyl N23 T24 : M	DPP9 [N2] / dpp9-xp-xa K2 F3 : N
CPB2 (active assumption) [C1] / cpb2-basic T24 K25 : M	ERAP2 [N1] / erap2-basic K1 K2 : M
CPN1 [C1] / cpn-basic T24 K25 : M	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp K1 K2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

NME1 p.Gly24Arg

CeGaT | 13 aa | bonds 1-12 | segment 1/1

PPISSCDTRTMAN

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12
	P P	P I	I S	S S	S C	C D	D T	T R	R T	T M	M A	A N
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.012	0.012	0.038	0.016	0.015	0.049	0.130	0.246	0.422	0.303	0.751	0.556
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.003	0.009	0.022	0.033	0.019	0.118	0.183	0.157	0.370	0.243	0.697	0.578
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.043	0.034	0.718	0.065	0.070	0.096	0.086	0.140	0.455	0.063	0.889	0.532
NetChop 20S 3.0 In-vitro proteasome digestion model	0.056	0.024	0.032	0.366	0.714	0.546	0.693	0.351	0.778	0.158	0.976	0.754
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.122	0.748	0.039	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	M	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	M	N	N	N	N	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa no assessed bond : NA	FAP [N2] / fap-dipeptidyl P2 I3 : M
XPNPEP1 [N1] / app1-xp P1 P2 : M	ENPEP [N1] / enpep-acidic P1 P2 : N
TPP2 [N3] / tpp2-tripeptidyl I3 S4 : M	ANPEP [N1] / anpep-ala P1 P2 : N
NPEPPS [N1] / npepps-n-terminal P1 P2 : N	DPP8 [N2] / dpp8-xp-xa P2 I3 : M
ACE [C2] / ace-dipeptidyl M11 A12 : M	DPP9 [N2] / dpp9-xp-xa P2 I3 : M
CPB2 (active assumption) [C1] / cpb2-basic A12 N13 : N	ERAP2 [N1] / erap2-basic P1 P2 : N
CPN1 [C1] / cpn-basic A12 N13 : N	ERAP1 [N1] / eramer-step P1 P2 : -0.011 PWM specificity
XPNPEP2 [N1] / app2-xp P1 P2 : M	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

NR2F2 p.Gly65Val

JLF V2 | 26 aa | bonds 1-24 | segment 1/2

AQTAAGGQGGPGVPGSDKQQQQQHI

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
	A Q	Q T	T A	A A	A G	G G	G Q	Q G	G G	G P	P G	G V	V P	P G	G S	S D	D K	K Q	Q Q	Q Q	Q Q	Q Q	Q H	H I
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.309	0.013	0.013	0.014	0.012	0.008	0.032	0.240	0.236	0.192	0.214	0.352	0.340	0.271	0.437	0.158	0.384	0.632	0.249	0.356	0.278	0.199	0.215	0.394
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.014	0.003	0.005	0.006	0.008	0.006	0.063	0.160	0.219	0.111	0.196	0.204	0.279	0.227	0.304	0.201	0.345	0.490	0.193	0.238	0.243	0.214	0.184	0.397
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.525	0.062	0.564	0.896	0.815	0.113	0.052	0.039	0.029	0.033	0.029	0.029	0.436	0.076	0.108	0.053	0.169	0.964	0.223	0.398	0.110	0.150	0.235	0.312
NetChop 20S 3.0 In-vitro proteasome digestion model	0.385	0.313	0.372	0.894	0.221	0.082	0.168	0.130	0.084	0.199	0.031	0.026	0.484	0.035	0.041	0.031	0.618	0.046	0.321	0.503	0.616	0.940	0.697	0.224
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.210	0.345	0.277	0.204	0.106	0.638	0.156	0.351	0.216	0.201	0.642	0.066	0.135	0.139	0.105	0.181	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.060	0.277	0.710	0.094	0.405	0.332	0.334	0.288	0.334	0.382	0.238	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	M	N	N	M	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	M
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	M	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa Q2 T3 : 0.024 log2 depletion	FAP [N2] / fap-dipeptidyl Q2 T3 : N
XPNPEP1 [N1] / app1-xp A1 Q2 : N	ENPEP [N1] / enpep-acidic A1 Q2 : N
TPP2 [N3] / tpp2-tripeptidyl T3 A4 : M	ANPEP [N1] / anpep-ala A1 Q2 : M
NPEPPS [N1] / npepps-n-terminal A1 Q2 : M	DPP8 [N2] / dpp8-xp-xa Q2 T3 : N
ACE [C2] / ace-dipeptidyl H24 I25 : N	DPP9 [N2] / dpp9-xp-xa Q2 T3 : N
CPB2 (active assumption) [C1] / cpb2-basic I25 E26 : N	ERAP2 [N1] / erap2-basic A1 Q2 : N
CPN1 [C1] / cpn-basic I25 E26 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp A1 Q2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

NR2F2 p.Gly65Val

JLF V2 | 26 aa | bonds 25-25 | segment 2/2

IE

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	25
	I E
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.739
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.731
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.816
NetChop 20S 3.0 In-vitro proteasome digestion model	0.234
NetCleave-I HLA MHC-I C-terminal processing proxy	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N
MME / mme-hydrophobic (preferred)	N
FAP / fap-endo-gp (required)	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa Q2 T3 : 0.024 log2 depletion	FAP [N2] / fap-dipeptidyl Q2 T3 : N
XPNPEP1 [N1] / app1-xp A1 Q2 : N	ENPEP [N1] / enpep-acidic A1 Q2 : N
TPP2 [N3] / tpp2-tripeptidyl T3 A4 : M	ANPEP [N1] / anpep-ala A1 Q2 : M
NPEPPS [N1] / npepps-n-terminal A1 Q2 : M	DPP8 [N2] / dpp8-xp-xa Q2 T3 : N
ACE [C2] / ace-dipeptidyl H24 I25 : N	DPP9 [N2] / dpp9-xp-xa Q2 T3 : N
CPB2 (active assumption) [C1] / cpb2-basic I25 E26 : N	ERAP2 [N1] / erap2-basic A1 Q2 : N
CPN1 [C1] / cpn-basic I25 E26 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp A1 Q2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

NR2F2 p.Gly65Val

CeGaT | 17 aa | bonds 1-16 | segment 1/1

AGGQGGPGVPGSDKQQQ

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	A G	G G	G Q	Q G	G G	G P	P G	G V	V P	P G	G S	S D	D K	K Q	Q Q	Q Q
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.413	0.007	0.015	0.022	0.005	0.023	0.021	0.272	0.340	0.261	0.399	0.236	0.363	0.634	0.241	0.486
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.007	0.002	0.006	0.012	0.006	0.012	0.015	0.134	0.279	0.189	0.314	0.197	0.342	0.619	0.285	0.556
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.567	0.063	0.093	0.063	0.031	0.043	0.028	0.027	0.436	0.090	0.145	0.057	0.170	0.962	0.410	0.482
NetChop 20S 3.0 In-vitro proteasome digestion model	0.231	0.080	0.148	0.130	0.084	0.199	0.031	0.026	0.484	0.035	0.041	0.031	0.618	0.046	0.402	0.562
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.106	0.638	0.156	0.351	0.216	0.201	0.642	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.405	0.332	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	M	N	N	M	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	M	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa G2 G3 : 0.193 log2 depletion	FAP [N2] / fap-dipeptidyl G2 G3 : N
XPNPEP1 [N1] / app1-xp A1 G2 : N	ENPEP [N1] / enpep-acidic A1 G2 : N
TPP2 [N3] / tpp2-tripeptidyl G3 Q4 : M	ANPEP [N1] / anpep-ala A1 G2 : M
NPEPPS [N1] / npepps-n-terminal A1 G2 : N	DPP8 [N2] / dpp8-xp-xa G2 G3 : N
ACE [C2] / ace-dipeptidyl Q15 Q16 : M	DPP9 [N2] / dpp9-xp-xa G2 G3 : N
CPB2 (active assumption) [C1] / cpb2-basic Q16 Q17 : N	ERAP2 [N1] / erap2-basic A1 G2 : N
CPN1 [C1] / cpn-basic Q16 Q17 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp A1 G2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

PIP5K1A p.Gly474fs

JLF V3 | 20 aa | bonds 1-19 | segment 1/1

SSFSRRAAPVATPALLTSHR

Source minimal epitope: AAPVATPAL (SLP 7-15)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
	S S	S F	F S	S R	R R	R A	A A	A P	P V	V A	A T	T P	P A	A L	L L	L T	T S	S H	H R
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.006	0.009	0.098	0.015	0.033	0.018	0.011	0.136	0.081	0.128	0.088	0.099	0.178	0.469	0.818	0.836	0.341	0.535	0.949
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.003	0.004	0.029	0.012	0.029	0.009	0.004	0.105	0.061	0.072	0.086	0.056	0.172	0.355	0.802	0.736	0.301	0.457	0.922
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.121	0.048	0.896	0.035	0.956	0.554	0.228	0.030	0.033	0.817	0.050	0.052	0.190	0.700	0.971	0.972	0.192	0.438	0.495
NetChop 20S 3.0 In-vitro proteasome digestion model	0.423	0.098	0.682	0.136	0.922	0.807	0.745	0.504	0.036	0.059	0.262	0.678	0.243	0.043	0.594	0.903	0.898	0.427	0.660
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.338	0.116	0.229	0.446	0.200	0.362	0.211	0.417	0.550	0.073	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.222	0.409	0.335	0.264	0.348	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	M	N	N	N	M	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	M	N	N	N	N	N	N	N	N	N	N	N	M	M	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa | S2|F3 : 0.616 log2 depletion

XPNPEP1 [N1] / app1-xp | S1|S2 : N

TPP2 [N3] / tpp2-tripeptidyl | F3|S4 : M

NPEPPS [N1] / npepps-n-terminal | S1|S2 : M

ACE [C2] / ace-dipeptidyl | S18|H19 : M

CPB2 (active assumption) [C1] / cpb2-basic | H19|R20 : M

CPN1 [C1] / cpn-basic | H19|R20 : M

XPNPEP2 [N1] / app2-xp | S1|S2 : N

FAP [N2] / fap-dipeptidyl | S2|F3 : N

ENPEP [N1] / enpep-acidic | S1|S2 : N

ANPEP [N1] / anpep-ala | S1|S2 : N

DPP8 [N2] / dpp8-xp-xa | S2|F3 : N

DPP9 [N2] / dpp9-xp-xa | S2|F3 : N

ERAP2 [N1] / erap2-basic | S1|S2 : N

ERAP1 [N1] / eramer-step | no assessed bond : NA

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

PTH1R p.Ile237Phe

JLF V3 | 23 aa | bonds 1-22 | segment 1/1

SFMLRAVSFFVKDAVLYSGAKKK

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
	S F	F M	M L	L R	R A	A V	V S	S F	F F	F V	V K	K D	D A	A V	V L	L Y	Y S	S G	G A	A K	K K	K K
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.004	0.019	0.025	0.046	0.020	0.009	0.030	0.080	0.842	0.650	0.364	0.344	0.104	0.121	0.186	0.545	0.929	0.237	0.189	0.366	0.827	0.931
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.002	0.014	0.009	0.028	0.014	0.006	0.018	0.044	0.799	0.513	0.328	0.407	0.113	0.127	0.196	0.638	0.840	0.175	0.124	0.367	0.770	0.902
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.024	0.602	0.150	0.900	0.228	0.050	0.119	0.029	0.271	0.214	0.474	0.948	0.071	0.271	0.186	0.947	0.936	0.089	0.195	0.198	0.839	0.859
NetChop 20S 3.0 In-vitro proteasome digestion model	0.272	0.907	0.215	0.962	0.952	0.609	0.960	0.514	0.682	0.109	0.874	0.172	0.968	0.763	0.389	0.471	0.953	0.136	0.057	0.053	0.031	0.147
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.028	0.583	0.382	0.184	0.375	0.111	0.362	0.164	0.148	0.745	0.020	0.184	0.271	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.411	0.291	0.260	0.408	0.213	0.257	0.316	0.285	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	M	N	M	N	N	N	N	M	M	N	N	N	N	N	M	M	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa F2 M3 : 0.012 log2 depletion	FAP [N2] / fap-dipeptidyl F2 M3 : N
XPNPEP1 [N1] / app1-xp S1 F2 : N	ENPEP [N1] / enpep-acidic S1 F2 : N
TPP2 [N3] / tpp2-tripeptidyl M3 L4 : M	ANPEP [N1] / anpep-ala S1 F2 : N
NPEPPS [N1] / npepps-n-terminal S1 F2 : M	DPP8 [N2] / dpp8-xp-xa F2 M3 : N
ACE [C2] / ace-dipeptidyl K21 K22 : M	DPP9 [N2] / dpp9-xp-xa F2 M3 : N
CPB2 (active assumption) [C1] / cpb2-basic K22 K23 : M	ERAP2 [N1] / erap2-basic S1 F2 : N
CPN1 [C1] / cpn-basic K22 K23 : M	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp S1 F2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

SLC25A12 p.Ile347Thr

JLF V2;JLF V1 | 26 aa | bonds 1-24 | segment 1/2

GSVAGAVGATAVYPTDLVKTRMQNQ

Source minimal epitope: ATAVYPTDL (SLP 9-17)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
	G S	S V	V A	A G	G A	A V	V G	G A	A T	T A	A V	V Y	Y P	P T	T D	D L	L V	V K	K T	T R	R M	M Q	Q N	N Q
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.004	0.011	0.012	0.010	0.008	0.022	0.044	0.144	0.407	0.304	0.380	0.431	0.700	0.113	0.034	0.099	0.340	0.313	0.751	0.641	0.913	0.839	0.216	0.230
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.002	0.003	0.004	0.003	0.005	0.021	0.041	0.086	0.281	0.119	0.361	0.451	0.738	0.087	0.037	0.116	0.273	0.358	0.749	0.594	0.865	0.695	0.268	0.326
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.035	0.024	0.786	0.163	0.056	0.103	0.169	0.033	0.138	0.057	0.208	0.265	0.970	0.023	0.033	0.046	0.756	0.541	0.939	0.184	0.944	0.957	0.088	0.347
NetChop 20S 3.0 In-vitro proteasome digestion model	0.117	0.094	0.400	0.275	0.531	0.227	0.079	0.389	0.810	0.299	0.638	0.380	0.915	0.456	0.039	0.356	0.193	0.751	0.261	0.536	0.956	0.852	0.882	0.478
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.131	0.764	0.091	0.325	0.229	0.761	0.069	0.195	0.116	0.484	0.215	0.359	0.159	0.493	0.348	0.054	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.265	0.188	0.473	0.297	0.453	0.220	0.513	0.337	0.433	0.205	0.429	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	M	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	N	N	N	N	N	N	N	N	M	N	N	N	M	N	N	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa S2 V3 : -0.580 log2 depletion	FAP [N2] / fap-dipeptidyl S2 V3 : N
XPNPEP1 [N1] / app1-xp G1 S2 : N	ENPEP [N1] / enpep-acidic G1 S2 : N
TPP2 [N3] / tpp2-tripeptidyl V3 A4 : M	ANPEP [N1] / anpep-ala G1 S2 : N
NPEPPS [N1] / npepps-n-terminal G1 S2 : N	DPP8 [N2] / dpp8-xp-xa S2 V3 : N
ACE [C2] / ace-dipeptidyl N24 Q25 : M	DPP9 [N2] / dpp9-xp-xa S2 V3 : N
CPB2 (active assumption) [C1] / cpb2-basic Q25 R26 : M	ERAP2 [N1] / erap2-basic G1 S2 : N
CPN1 [C1] / cpn-basic Q25 R26 : M	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp G1 S2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

SLC25A12 p.Ile347Thr

JLF V2;JLF V1 | 26 aa | bonds 25-25 | segment 2/2

QR

Source minimal epitope: ATAVYPTDL (SLP 9-17)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	25
	Q R
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.606
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.646
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.192
NetChop 20S 3.0 In-vitro proteasome digestion model	0.828
NetCleave-I HLA MHC-I C-terminal processing proxy	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N
MME / mme-hydrophobic (preferred)	N
FAP / fap-endo-gp (required)	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa S2 V3 : -0.580 log2 depletion	FAP [N2] / fap-dipeptidyl S2 V3 : N
XPNPEP1 [N1] / app1-xp G1 S2 : N	ENPEP [N1] / enpep-acidic G1 S2 : N
TPP2 [N3] / tpp2-tripeptidyl V3 A4 : M	ANPEP [N1] / anpep-ala G1 S2 : N
NPEPPS [N1] / npepps-n-terminal G1 S2 : N	DPP8 [N2] / dpp8-xp-xa S2 V3 : N
ACE [C2] / ace-dipeptidyl N24 Q25 : M	DPP9 [N2] / dpp9-xp-xa S2 V3 : N
CPB2 (active assumption) [C1] / cpb2-basic Q25 R26 : M	ERAP2 [N1] / erap2-basic G1 S2 : N
CPN1 [C1] / cpn-basic Q25 R26 : M	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp G1 S2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

SLC25A12 p.Ile347Thr

CeGaT | 17 aa | bonds 1-16 | segment 1/1

VGATAVYPTDLVKTRMQ

Source minimal epitope: ATAVYPTDL (SLP 3-11)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	V G	G A	A T	T A	A V	V Y	Y P	P T	T D	D L	L V	V K	K T	T R	R M	M Q
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.102	0.012	0.010	0.013	0.015	0.023	0.147	0.069	0.034	0.223	0.312	0.358	0.798	0.608	0.958	0.929
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.081	0.001	0.003	0.007	0.008	0.011	0.151	0.048	0.037	0.146	0.294	0.314	0.702	0.603	0.888	0.885
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.434	0.033	0.137	0.037	0.481	0.465	0.948	0.023	0.033	0.048	0.832	0.572	0.933	0.402	0.951	0.970
NetChop 20S 3.0 In-vitro proteasome digestion model	0.209	0.363	0.648	0.299	0.638	0.380	0.915	0.456	0.039	0.356	0.193	0.751	0.261	0.536	0.949	0.635
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.069	0.195	0.116	0.484	0.215	0.359	0.159	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.513	0.337	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	M	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	N	N	M	N	N	N	M	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa G2 A3 : 0.686 log2 depletion	FAP [N2] / fap-dipeptidyl G2 A3 : N
XPNPEP1 [N1] / app1-xp V1 G2 : N	ENPEP [N1] / enpep-acidic V1 G2 : N
TPP2 [N3] / tpp2-tripeptidyl A3 T4 : M	ANPEP [N1] / anpep-ala V1 G2 : N
NPEPPS [N1] / npepps-n-terminal V1 G2 : N	DPP8 [N2] / dpp8-xp-xa G2 A3 : N
ACE [C2] / ace-dipeptidyl R15 M16 : M	DPP9 [N2] / dpp9-xp-xa G2 A3 : N
CPB2 (active assumption) [C1] / cpb2-basic M16 Q17 : N	ERAP2 [N1] / erap2-basic V1 G2 : N
CPN1 [C1] / cpn-basic M16 Q17 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp V1 G2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

SMC5 p.Arg371Lys

JLF V3 | 21 aa | bonds 1-20 | segment 1/1

DRQKRIGNTRKMIEDLQNELK

Source minimal epitope: RQKRIGNTR (SLP 2-10)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

Pepsickle human [preferred]

MHC-I epitope proxy; proteasome type agnostic

Pepsickle all-mammal [comparison]

MHC-I epitope proxy; proteasome type agnostic

NetChop Cterm 3.0

MHC-I ligand C-terminal processing proxy

NetChop 20S 3.0

In-vitro proteasome digestion model

NetCleave-I HLA

MHC-I C-terminal processing proxy

NetCleave-II HLA

MHC-II processing proxy; no named cathepsin

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
D R	R Q	Q K	K R	R I	I G	G N	N T	T R	R K	K M	M I	I E	E D	D L	L Q	Q N	N E	E L	L K
0.033	0.015	0.008	0.009	0.040	0.024	0.047	0.183	0.169	0.709	0.489	0.725	0.260	0.107	0.111	0.289	0.107	0.125	0.305	0.974
0.109	0.003	0.006	0.009	0.041	0.019	0.025	0.084	0.097	0.656	0.477	0.755	0.324	0.069	0.071	0.260	0.087	0.145	0.342	0.947
0.030	0.306	0.028	0.309	0.147	0.285	0.025	0.024	0.039	0.644	0.820	0.941	0.359	0.024	0.028	0.712	0.062	0.038	0.071	0.974
0.447	0.162	0.405	0.491	0.809	0.780	0.514	0.221	0.078	0.941	0.325	0.567	0.953	0.594	0.783	0.791	0.326	0.299	0.098	0.620
NA	NA	NA	NA	NA	NA	NA	0.242	0.098	0.736	0.213	0.360	0.293	0.074	0.100	0.891	0.062	0.259	NA	NA
NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.176	0.374	0.360	0.282	0.297	0.366	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)

MME / mme-hydrophobic (preferred)

FAP / fap-endo-gp (required)

N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
N	N	N	N	M	N	N	N	N	N	N	M	N	N	M	N	N	N	M	N
N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa | no assessed bond : NA

XPNPEP1 [N1] / app1-xp | D1|R2 : N

TPP2 [N3] / tpp2-tripeptidyl | Q3|K4 : M

NPEPPS [N1] / npepps-n-terminal | D1|R2 : M

ACE [C2] / ace-dipeptidyl | E19|L20 : M

CPB2 (active assumption) [C1] / cpb2-basic | L20|K21 : M

CPN1 [C1] / cpn-basic | L20|K21 : M

XPNPEP2 [N1] / app2-xp | D1|R2 : N

FAP [N2] / fap-dipeptidyl | R2|Q3 : N

ENPEP [N1] / enpep-acidic | D1|R2 : M

ANPEP [N1] / anpep-ala | D1|R2 : N

DPP8 [N2] / dpp8-xp-xa | R2|Q3 : N

DPP9 [N2] / dpp9-xp-xa | R2|Q3 : N

ERAP2 [N1] / erap2-basic | D1|R2 : N

ERAP1 [N1] / eramer-step | no assessed bond : NA

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

TECPR1 p.Thr259fs

JLF V3;JLF V2;JLF V1 | 22 aa | bonds 1-21 | segment 1/1

GSEESTLTIPAAPVGLRWLVR

Source minimal epitope: TIPAAPVGL (SLP 9-17)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
	G S	S E	E E	E A	A S	S T	T L	L T	T I	I P	P A	A A	A P	P V	V G	G L	L R	R W	W L	L V	V R
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.007	0.008	0.011	0.011	0.008	0.006	0.035	0.337	0.166	0.694	0.291	0.294	0.128	0.076	0.157	0.152	0.715	0.750	0.908	0.804	0.939
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.008	0.003	0.009	0.012	0.006	0.012	0.021	0.164	0.104	0.716	0.215	0.161	0.087	0.059	0.107	0.193	0.591	0.640	0.866	0.767	0.918
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.038	0.050	0.047	0.042	0.675	0.027	0.027	0.800	0.066	0.860	0.025	0.258	0.129	0.027	0.750	0.048	0.613	0.655	0.945	0.734	0.960
NetChop 20S 3.0 In-vitro proteasome digestion model	0.102	0.078	0.198	0.373	0.881	0.125	0.150	0.896	0.059	0.758	0.063	0.056	0.425	0.100	0.043	0.047	0.991	0.616	0.921	0.395	0.926
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.916	0.022	0.895	0.071	0.328	0.600	0.190	0.541	0.044	0.576	0.164	0.720	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.092	0.264	0.350	0.378	0.235	0.378	0.331	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	M	N	N	M	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	N	N	N	M	N	M	N	N	N	N	N	N	M	N	N	M	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa | S2|E3 : -0.085 log2 depletion

XPNPEP1 [N1] / app1-xp | G1|S2 : N

TPP2 [N3] / tpp2-tripeptidyl | E3|E4 : M

NPEPPS [N1] / npepps-n-terminal | G1|S2 : N

ACE [C2] / ace-dipeptidyl | L20|V21 : M

CPB2 (active assumption) [C1] / cpb2-basic | V21|R22 : M

CPN1 [C1] / cpn-basic | V21|R22 : M

XPNPEP2 [N1] / app2-xp | G1|S2 : N

FAP [N2] / fap-dipeptidyl | S2|E3 : N

ENPEP [N1] / enpep-acidic | G1|S2 : N

ANPEP [N1] / anpep-ala | G1|S2 : N

DPP8 [N2] / dpp8-xp-xa | S2|E3 : N

DPP9 [N2] / dpp9-xp-xa | S2|E3 : N

ERAP2 [N1] / erap2-basic | G1|S2 : N

ERAP1 [N1] / eramer-step | no assessed bond : NA

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

TECPR1 p.Thr259fs

JLF V2;JLF V1 | 17 aa | bonds 1-16 | segment 1/1

TSRWESAWSGLSPRTGR

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	T S	S R	R W	W E	E S	S A	A W	W S	S G	G L	L S	S P	P R	R T	T G	G R
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.019	0.009	0.013	0.117	0.009	0.005	0.033	0.692	0.062	0.137	0.628	0.193	0.303	0.685	0.326	0.590
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.003	0.006	0.010	0.108	0.009	0.006	0.028	0.644	0.068	0.102	0.591	0.113	0.267	0.640	0.206	0.584
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.076	0.040	0.773	0.966	0.031	0.075	0.090	0.867	0.065	0.569	0.658	0.041	0.068	0.792	0.316	0.065
NetChop 20S 3.0 In-vitro proteasome digestion model	0.060	0.667	0.567	0.415	0.162	0.926	0.447	0.835	0.079	0.131	0.836	0.059	0.376	0.127	0.059	0.787
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.929	0.009	0.059	0.803	0.048	0.211	0.668	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.299	0.498	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	M	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	N	N	N	N	N	N	M	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa no assessed bond : NA	FAP [N2] / fap-dipeptidyl S2 R3 : N
XPNPEP1 [N1] / app1-xp T1 S2 : N	ENPEP [N1] / enpep-acidic T1 S2 : N
TPP2 [N3] / tpp2-tripeptidyl R3 W4 : M	ANPEP [N1] / anpep-ala T1 S2 : N
NPEPPS [N1] / npepps-n-terminal T1 S2 : M	DPP8 [N2] / dpp8-xp-xa S2 R3 : N
ACE [C2] / ace-dipeptidyl T15 G16 : M	DPP9 [N2] / dpp9-xp-xa S2 R3 : N
CPB2 (active assumption) [C1] / cpb2-basic G16 R17 : M	ERAP2 [N1] / erap2-basic T1 S2 : N
CPN1 [C1] / cpn-basic G16 R17 : M	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp T1 S2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

TECPR1 p.Thr259fs

CeGaT | 16 aa | bonds 1-15 | segment 1/1

LWAHSGRDRPWSGKES

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	L W	W A	A H	H S	S G	G R	R D	D R	R P	P W	W S	S G	G K	K E	E S
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.351	0.174	0.018	0.020	0.015	0.008	0.031	0.082	0.250	0.153	0.408	0.203	0.197	0.669	0.446
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.028	0.063	0.011	0.014	0.008	0.014	0.036	0.134	0.223	0.109	0.416	0.169	0.206	0.734	0.406
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.749	0.907	0.204	0.444	0.092	0.233	0.501	0.189	0.037	0.034	0.805	0.023	0.320	0.971	0.257
NetChop 20S 3.0 In-vitro proteasome digestion model	0.493	0.942	0.205	0.914	0.080	0.595	0.204	0.134	0.942	0.048	0.057	0.030	0.678	0.064	0.415
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.063	0.801	0.037	0.778	0.026	0.143	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.351	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	M	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa W2 A3 : -0.010 log2 depletion	FAP [N2] / fap-dipeptidyl W2 A3 : N
XPNPEP1 [N1] / app1-xp L1 W2 : N	ENPEP [N1] / enpep-acidic L1 W2 : N
TPP2 [N3] / tpp2-tripeptidyl A3 H4 : M	ANPEP [N1] / anpep-ala L1 W2 : N
NPEPPS [N1] / npepps-n-terminal L1 W2 : M	DPP8 [N2] / dpp8-xp-xa W2 A3 : N
ACE [C2] / ace-dipeptidyl K14 E15 : M	DPP9 [N2] / dpp9-xp-xa W2 A3 : N
CPB2 (active assumption) [C1] / cpb2-basic E15 S16 : N	ERAP2 [N1] / erap2-basic L1 W2 : N
CPN1 [C1] / cpn-basic E15 S16 : N	ERAP1 [N1] / eramer-step L1 W2 : -0.182 PWM specificity
XPNPEP2 [N1] / app2-xp L1 W2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

VPS13B p.Ser3756Phe

JLF V3 | 21 aa | bonds 1-20 | segment 1/1

RQLPESLGEGLRQGLFRLGIS

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
	R Q	Q L	L P	P E	E S	S L	L G	G E	E G	G L	L R	R Q	Q G	G L	L F	F R	R L	L G	G I	I S
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.062	0.008	0.012	0.004	0.013	0.007	0.051	0.130	0.128	0.194	0.784	0.264	0.081	0.184	0.578	0.702	0.547	0.712	0.345	0.882
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.004	0.004	0.003	0.005	0.037	0.010	0.249	0.081	0.117	0.118	0.719	0.250	0.114	0.120	0.539	0.590	0.465	0.620	0.305	0.896
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.653	0.060	0.889	0.051	0.031	0.037	0.849	0.090	0.039	0.078	0.739	0.451	0.052	0.118	0.491	0.388	0.064	0.976	0.213	0.575
NetChop 20S 3.0 In-vitro proteasome digestion model	0.502	0.269	0.690	0.049	0.106	0.167	0.756	0.149	0.765	0.051	0.834	0.638	0.811	0.787	0.653	0.800	0.957	0.872	0.084	0.815
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.032	0.342	0.119	0.616	0.158	0.093	0.063	0.409	0.485	0.166	0.661	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.282	0.498	0.299	0.282	0.365	0.253	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	M	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	M	N	N	N	M	N	N	N	M	N	N	N	M	M	N	M	N	M	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa | Q2|L3 : 0.023 log2 depletion

XPNPEP1 [N1] / app1-xp | R1|Q2 : N

TPP2 [N3] / tpp2-tripeptidyl | L3|P4 : N

NPEPPS [N1] / npepps-n-terminal | R1|Q2 : M

ACE [C2] / ace-dipeptidyl | G19|I20 : M

CPB2 (active assumption) [C1] / cpb2-basic | I20|S21 : N

CPN1 [C1] / cpn-basic | I20|S21 : N

XPNPEP2 [N1] / app2-xp | R1|Q2 : N

FAP [N2] / fap-dipeptidyl | Q2|L3 : N

ENPEP [N1] / enpep-acidic | R1|Q2 : N

ANPEP [N1] / anpep-ala | R1|Q2 : N

DPP8 [N2] / dpp8-xp-xa | Q2|L3 : N

DPP9 [N2] / dpp9-xp-xa | Q2|L3 : N

ERAP2 [N1] / erap2-basic | R1|Q2 : M

ERAP1 [N1] / eramer-step | no assessed bond : NA

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

VPS13B p.Ser3756Phe

CeGaT | 20 aa | bonds 1-19 | segment 1/1

GEGLRQGLFRLGISLLGAIA

No located source minimal epitope for this SLP; all internal bonds still shown

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
	G E	E G	G L	L R	R Q	Q G	G L	L F	F R	R L	L G	G I	I S	S L	L L	L G	G A	A I	I A
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.009	0.037	0.006	0.019	0.028	0.027	0.010	0.404	0.687	0.432	0.558	0.151	0.678	0.159	0.806	0.538	0.208	0.713	0.793
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.011	0.013	0.003	0.024	0.026	0.013	0.025	0.235	0.453	0.404	0.591	0.126	0.709	0.091	0.640	0.605	0.210	0.637	0.708
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.045	0.028	0.032	0.561	0.424	0.047	0.041	0.381	0.364	0.048	0.976	0.041	0.196	0.027	0.917	0.877	0.493	0.337	0.289
NetChop 20S 3.0 In-vitro proteasome digestion model	0.316	0.429	0.048	0.834	0.638	0.811	0.787	0.653	0.800	0.957	0.872	0.145	0.960	0.138	0.894	0.586	0.507	0.352	0.295
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.409	0.485	0.166	0.661	0.041	0.802	0.049	0.614	0.258	0.049	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.166	0.331	0.323	0.317	0.356	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	M	N	N	N	M	M	N	M	N	M	N	M	M	N	N	M	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa | E2|G3 : 0.020 log2 depletion

XPNPEP1 [N1] / app1-xp | G1|E2 : N

TPP2 [N3] / tpp2-tripeptidyl | G3|L4 : M

NPEPPS [N1] / npepps-n-terminal | G1|E2 : N

ACE [C2] / ace-dipeptidyl | A18|I19 : M

CPB2 (active assumption) [C1] / cpb2-basic | I19|A20 : N

CPN1 [C1] / cpn-basic | I19|A20 : N

XPNPEP2 [N1] / app2-xp | G1|E2 : N

FAP [N2] / fap-dipeptidyl | E2|G3 : N

ENPEP [N1] / enpep-acidic | G1|E2 : N

ANPEP [N1] / anpep-ala | G1|E2 : N

DPP8 [N2] / dpp8-xp-xa | E2|G3 : N

DPP9 [N2] / dpp9-xp-xa | E2|G3 : N

ERAP2 [N1] / erap2-basic | G1|E2 : N

ERAP1 [N1] / eramer-step | no assessed bond : NA

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

VPS72 p.Lys115Arg

JLF V3 | 27 aa | bonds 1-24 | segment 1/2

GSSQKAREERALLPLELQDDGSDSR

Source minimal epitope: ERALLPLEL (SLP 9-17)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
	G S	S S	S Q	Q K	K A	A R	R E	E E	E R	R A	A L	L L	L P	P L	L E	E L	L Q	Q D	D D	D G	G S	S D	D S	S R
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.010	0.007	0.011	0.016	0.024	0.012	0.035	0.109	0.109	0.298	0.081	0.268	0.533	0.134	0.469	0.340	0.868	0.151	0.205	0.289	0.149	0.104	0.288	0.220
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.046	0.003	0.006	0.011	0.013	0.011	0.030	0.070	0.142	0.291	0.077	0.262	0.537	0.109	0.448	0.210	0.894	0.174	0.115	0.192	0.101	0.078	0.198	0.312
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.045	0.039	0.052	0.148	0.967	0.659	0.179	0.036	0.033	0.284	0.172	0.093	0.915	0.329	0.972	0.030	0.971	0.078	0.214	0.092	0.161	0.026	0.056	0.234
NetChop 20S 3.0 In-vitro proteasome digestion model	0.055	0.123	0.032	0.509	0.227	0.520	0.762	0.234	0.912	0.960	0.217	0.553	0.792	0.073	0.143	0.178	0.968	0.391	0.785	0.078	0.099	0.039	0.109	0.043
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.037	0.164	0.585	0.081	0.346	0.520	0.121	0.639	0.031	0.954	0.027	0.234	0.388	0.316	0.170	0.549	0.110
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.251	0.232	0.432	0.319	0.406	0.353	0.164	0.355	0.508	0.194	0.327	0.305

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	M	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	N	N	N	N	N	N	N	N	M	M	N	M	N	M	N	N	N	N	N	N	N	N
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa S2 S3 : 0.343 log2 depletion	FAP [N2] / fap-dipeptidyl S2 S3 : N
XPNPEP1 [N1] / app1-xp G1 S2 : N	ENPEP [N1] / enpep-acidic G1 S2 : N
TPP2 [N3] / tpp2-tripeptidyl S3 Q4 : M	ANPEP [N1] / anpep-ala G1 S2 : N
NPEPPS [N1] / npepps-n-terminal G1 S2 : N	DPP8 [N2] / dpp8-xp-xa S2 S3 : N
ACE [C2] / ace-dipeptidyl R25 K26 : M	DPP9 [N2] / dpp9-xp-xa S2 S3 : N
CPB2 (active assumption) [C1] / cpb2-basic K26 S27 : N	ERAP2 [N1] / erap2-basic G1 S2 : N
CPN1 [C1] / cpn-basic K26 S27 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp G1 S2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

VPS72 p.Lys115Arg

JLF V3 | 27 aa | bonds 25-26 | segment 2/2

RKS

Source minimal epitope: ERALLPLEL (SLP 9-17)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	25	26
	R K	K S
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.801	0.732
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.748	0.762
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.902	0.898
NetChop 20S 3.0 In-vitro proteasome digestion model	0.625	0.065
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N
MME / mme-hydrophobic (preferred)	N	N
FAP / fap-endo-gp (required)	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa S2 S3 : 0.343 log2 depletion	FAP [N2] / fap-dipeptidyl S2 S3 : N
XPNPEP1 [N1] / app1-xp G1 S2 : N	ENPEP [N1] / enpep-acidic G1 S2 : N
TPP2 [N3] / tpp2-tripeptidyl S3 Q4 : M	ANPEP [N1] / anpep-ala G1 S2 : N
NPEPPS [N1] / npepps-n-terminal G1 S2 : N	DPP8 [N2] / dpp8-xp-xa S2 S3 : N
ACE [C2] / ace-dipeptidyl R25 K26 : M	DPP9 [N2] / dpp9-xp-xa S2 S3 : N
CPB2 (active assumption) [C1] / cpb2-basic K26 S27 : N	ERAP2 [N1] / erap2-basic G1 S2 : N
CPN1 [C1] / cpn-basic K26 S27 : N	ERAP1 [N1] / eramer-step no assessed bond : NA
XPNPEP2 [N1] / app2-xp G1 S2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

WWC1 p.Val179Gly

JLF V3 | 20 aa | bonds 1-19 | segment 1/1

NKCLKREMGHLQHELQFKERG

Source minimal epitope: GHLQHELQF (SLP 8-16)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

Pepsickle human [preferred]

MHC-I epitope proxy; proteasome type agnostic

Pepsickle all-mammal [comparison]

MHC-I epitope proxy; proteasome type agnostic

NetChop Cterm 3.0

MHC-I ligand C-terminal processing proxy

NetChop 20S 3.0

In-vitro proteasome digestion model

NetCleave-I HLA

MHC-I C-terminal processing proxy

NetCleave-II HLA

MHC-II processing proxy; no named cathepsin

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
N K	K L	L K	K R	R E	E M	M G	G H	H L	L Q	Q H	H E	E L	L Q	Q F	F K	K E	E R	R G
0.021	0.016	0.029	0.029	0.011	0.010	0.139	0.036	0.069	0.187	0.046	0.110	0.065	0.696	0.203	0.864	0.680	0.294	0.889
0.004	0.008	0.022	0.020	0.016	0.015	0.085	0.062	0.071	0.188	0.047	0.234	0.094	0.608	0.210	0.817	0.709	0.364	0.825
0.054	0.371	0.942	0.404	0.676	0.046	0.185	0.030	0.032	0.906	0.035	0.683	0.036	0.971	0.042	0.925	0.682	0.169	0.620
0.128	0.047	0.494	0.218	0.878	0.954	0.958	0.312	0.361	0.848	0.564	0.607	0.341	0.820	0.045	0.911	0.129	0.858	0.777
NA	NA	NA	NA	NA	NA	NA	0.078	0.173	0.608	0.100	0.616	0.015	0.946	0.021	0.879	0.106	NA	NA
NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.444	0.289	0.365	0.350	0.387	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)

MME / mme-hydrophobic (preferred)

FAP / fap-endo-gp (required)

N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
N	M	N	N	N	N	N	N	M	N	N	N	M	N	M	N	N	N	N
N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa | K2|L3 : 0.065 log2 depletion

XPNPEP1 [N1] / app1-xp | N1|K2 : N

TPP2 [N3] / tpp2-tripeptidyl | L3|K4 : M

NPEPPS [N1] / npepps-n-terminal | N1|K2 : M

ACE [C2] / ace-dipeptidyl | E18|R19 : M

CPB2 (active assumption) [C1] / cpb2-basic | R19|G20 : N

CPN1 [C1] / cpn-basic | R19|G20 : N

XPNPEP2 [N1] / app2-xp | N1|K2 : N

FAP [N2] / fap-dipeptidyl | K2|L3 : N

ENPEP [N1] / enpep-acidic | N1|K2 : N

ANPEP [N1] / anpep-ala | N1|K2 : N

DPP8 [N2] / dpp8-xp-xa | K2|L3 : N

DPP9 [N2] / dpp9-xp-xa | K2|L3 : N

ERAP2 [N1] / erap2-basic | N1|K2 : N

ERAP1 [N1] / eramer-step | no assessed bond : NA

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.

ZNF436 p.Ser207Cys

CeGaT | 16 aa | bonds 1-15 | segment 1/1

GKSFGRSCHLIQHQT**I**

Source minimal epitope: KSFGRSCHL (SLP 2-10)

INTERNAL SITE SCORES - native 0-1 outputs; bold cells meet that model's 0.5 display threshold

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	G K	K S	S F	F G	G R	R S	S C	C H	H L	L I	I Q	Q H	H Q	Q T	T I
Pepsickle human [preferred] MHC-I epitope proxy; proteasome type agnostic	0.011	0.017	0.009	0.051	0.004	0.028	0.006	0.117	0.120	0.522	0.428	0.174	0.364	0.265	0.354
Pepsickle all-mammal [comparison] MHC-I epitope proxy; proteasome type agnostic	0.036	0.003	0.006	0.023	0.020	0.027	0.062	0.112	0.071	0.437	0.432	0.185	0.646	0.264	0.415
NetChop Cterm 3.0 MHC-I ligand C-terminal processing proxy	0.059	0.560	0.054	0.623	0.044	0.661	0.032	0.026	0.124	0.925	0.596	0.034	0.406	0.070	0.169
NetChop 20S 3.0 In-vitro proteasome digestion model	0.071	0.414	0.130	0.740	0.084	0.956	0.647	0.843	0.437	0.627	0.725	0.495	0.915	0.729	0.208
NetCleave-I HLA MHC-I C-terminal processing proxy	NA	NA	NA	NA	NA	NA	NA	0.582	0.077	0.466	0.669	0.035	0.646	NA	NA
NetCleave-II HLA MHC-II processing proxy; no named cathepsin	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.380	NA	NA

INTERNAL ENZYME MOTIFS - M = matched; N = not matched; NA = outside scope (no numerical score)

PREP / prep-pro (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
MME / mme-hydrophobic (preferred)	N	N	M	N	N	N	N	N	M	M	N	N	N	N	M
FAP / fap-endo-gp (required)	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

EXPOSED TERMINI ONLY - N1/N2/N3 remove 1/2/3 N-terminal residues; C1/C2 remove 1/2 C-terminal residues

DPP4 [N2] / dpp4-qpisa no assessed bond : NA	FAP [N2] / fap-dipeptidyl K2 S3 : N
XPNPEP1 [N1] / app1-xp G1 K2 : N	ENPEP [N1] / enpep-acidic G1 K2 : N
TPP2 [N3] / tpp2-tripeptidyl S3 F4 : M	ANPEP [N1] / anpep-ala G1 K2 : N
NPEPPS [N1] / npepps-n-terminal G1 K2 : N	DPP8 [N2] / dpp8-xp-xa K2 S3 : N
ACE [C2] / ace-dipeptidyl Q14 T15 : M	DPP9 [N2] / dpp9-xp-xa K2 S3 : N
CPB2 (active assumption) [C1] / cpb2-basic T15 I16 : N	ERAP2 [N1] / erap2-basic G1 K2 : N
CPN1 [C1] / cpn-basic T15 I16 : N	ERAP1 [N1] / eramer-step G1 K2 : -0.061 PWM specificity
XPNPEP2 [N1] / app2-xp G1 K2 : N	

Gold bond headers are strictly inside the located source epitope. Exposure and chemical assumptions are in the catalog. CSV retains unassessed reasons and native precision. No score aggregation or resistance claim.